

REVIEW ARTICLE

Optical Microcavities Empowered Biochemical Sensing: Status and Prospects

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Optical microcavities are compact structures that confine resonant photons in microscale dimensions for long periods of time, greatly enhancing light–matter interactions. Plentiful and profound physical mechanisms within these microcavities or functional microcavities have been extensively explored, including mode shift/splitting/broadening, lasing and gain enhancements, surface plasmon resonance, fluorescence resonance energy transferring, optical frequency comb spectroscopy, optomechanical interaction, and exceptional point. The versatility in design and the diverse range of materials, particularly composites involving metals and 2-dimensional materials, have paved a way for innovative approaches and improved performance in biochemical sensing applications. Leveraging the advantages ranging from miniaturization, high sensitivity, rapid response, and inherent stability, optical microcavity-based biochemical sensors have emerged to address the growing and increasingly complex demands of biochemical detection. This review commences with an exploration of fundamental mechanisms and structures and then delves into typical applications in recent advancements, covering the detection of biomacromolecules, cells, solid particles, liquid ions, and gas molecules. This review also culminates with a forward-looking perspective, highlighting future development trends and crucial research directions.

Introduction

Optical sensors, with diverse optical transduction mechanisms, well-designed structures, and a range of material platforms, offer efficient means to accurately detect subtle environmental changes, comprehend dynamic molecular processes, and advance Internet of Things systems [1,2]. Their high precision, rapid response, inherent stability, versatility, and biocompatibility make optical sensors invaluable for applications in biochemical detection, environmental monitoring, clinical diagnostics, industrial production, and pharmaceutical research [3,4]. The increasing demand for biochemical detection necessitates optical sensors with ultrahigh sensitivity, real-time capabilities, miniaturization, cost-effectiveness, multicomponent functionality, and selectivity.

To better clarify these requirements, researchers have established quantitative parameters, including sensitivity, stability, detection limits, response/recovery times, multicomponent capabilities, and selectivity [1,5]. Traditional optical sensor devices often rely on optical fibers, incorporating structures such as D-shaped fibers, microfibers, multimode interferometers,

hollow-core fibers, and Bragg reflection fibers [6–8]. However, these sensors, despite improvements in performance [9], face limitations due to their interactions with biochemical molecules mainly through evanescent fields, ignoring the further enhancement in integrity, sensitivity, rapid detection, and selectivity. The flourishing development of micro/nano fabrication techniques has paved an avenue for geometric design and functional modification at the micro/nano scale [2]. Optical microcavities, characterized by high quality factor and small mode volume, have found applications in diverse fields, including microlasers, symmetric broken systems, optical frequency combs, quantum optics, optical calculations, and optical sensing [10–16].

In particular, in the realm of biochemical sensing, optical microcavities have become a vital research direction. Their ultrahigh quality factor and compact mode volume extend photon lifetimes and elevate the energy density of the optical field, resulting in strong light–matter interactions that enhance sensitivity and lower detection limits [17]. The rich underlying physical mechanisms of optical microcavities also provide powerful tools for enhancing sensing capabilities. Furthermore,

Citation: Guo Y, Liang Y, Li Y, Tian B, Fan X, He Y, Liu M, Peng L, Tang N, Tan T, et al. Optical Microcavities Empowered Biochemical Sensing: Status and Prospects. *Adv. Devices Instrum.* 2024;5:Article 0041. <https://doi.org/10.34133/adi.0041>

Submitted 22 October 2023
Accepted 22 December 2023
Published 20 March 2024

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optical microcavities possess characteristics such as compact size, stability, and good biocompatibility, making real-time tracking of single cell possible [18]. Furthermore, micro/nano engineering has facilitated the integration of gain and functional materials with optical microcavities. This integration empowers optical microcavities with the outstanding optoelectronic properties of materials such as graphene and transition metal dichalcogenides (TMDs), infusing them with boundless potential [19–21]. Novel principles and structural designs have given rise to a series of high-performance optical microcavities biochemical sensing devices [8,22,23].

Based on the substantial advantages of optical microcavities and the rapid developments in the field of biochemical sensing, this article presents a comprehensive review of optical microcavity-based biochemical sensors. It encompasses 4 sections: sensing mechanisms, structural designs, state-of-the-art biochemical sensors, and future prospects. To begin, the review delves into major microcavity parameters and elaborates on typical and impressive sensing mechanisms. These mechanisms include lasing and gain enhancements, surface plasmons, fluorescence resonance energy transfer, exceptional points (EPs), and more, as discussed in the mechanism section. Moving on, we provide in-depth insights into optical microcavity structures, materials, fabrication processes, and their combinations. Particular attention is given to functionalized materials and their fabrication methods. In the device section, the review explores recent advancements in the scenarios of detecting biomacromolecules, cells, solid particles, liquid ions, and gas molecules from biological and chemical perspectives. Finally, building upon current researches, the paper summarizes the existing challenges in the field of optical microcavity-based biochemical sensing and highlights future development trends.

The biochemical sensing mechanisms using optical microcavities

One of superiorities of microcavities is that they allow abundant physical phenomena. This section introduces optical microcavity-based biochemical sensing principles, starting with important parameters such as quality factor, mode volume, and finesse. Building on the foundation, nonlinear sensing principles, including optical frequency combs spectroscopy, laser sensing enhancement, optomechanical interaction, and photoacoustic and photothermal effect, are presented. Additionally, we demonstrate cutting-edge mechanisms, such as surface plasmon resonances (SPRs), fluorescence resonance energy transfer, and EPs for the applications of biochemical sensing.

Basic description of an optical microcavity

The quality factor (Q) is a primary parameter to describe microcavities and its temporal confinement of photon. When light propagates in microcavities, the intracavity energy W exhibits exponential decays, $W(t) = W_0 e^{-t/\tau}$, where W_0 and τ present the initial energy and photon lifetime, respectively. The quality factor Q is defined as the ratio of the intracavity stored energy to the energy-losing rate.

$$Q = \omega_m \frac{W}{-dW/dt} = \frac{\lambda_m}{\delta\lambda} = \omega_m \tau \quad (1)$$

where ω_m and λ_m are the resonance angular frequency and wavelength, respectively, and is the full width at half maximum (FWHM) of the resonance mode. The losses in microcavities mainly come from radiation ($1/Q_{\text{rad}}$), material ($1/Q_{\text{mat}}$), surface

and bulk scattering ($1/Q_{\text{sca}}$), and coupling ($1/Q_{\text{coup}}$). These losses impact the mode linewidth and result in total Q given by

$$Q^{-1} = Q_{\text{rad}}^{-1} + Q_{\text{mat}}^{-1} + Q_{\text{sca}}^{-1} + Q_{\text{coup}}^{-1} + \dots \quad (2)$$

Radiation loss is related to the microcavity size and is dominated in small resonators. As the resonator size increases, material absorption loss also increases, and the host medium becomes a limiting factor for the microcavity's Q , given by $Q_{\text{abs}} = 2\pi n_r / (\alpha \lambda_m)$, where n_r is the refractive index of the resonator, and α is the medium absorption coefficient. For instance, the ultimate material limited Q factor for a silica microsphere cavity is around 10^{10} [24].

The mode volume (V) is an important parameter to illustrate microcavities and its spatial confinement of photons, which is defined as

$$V = \frac{\int \rho dx dy dz}{\max(\rho)} \quad (3)$$

where ρ represents the energy density of the optical field, defined as $\rho = \epsilon E^2$, ϵ is the permittivity, and E is the electric field strength [25]. A smaller mode volume indicates stronger spatial confinement of light, resulting in enhanced light-matter interaction. The mode volume is closely related to the type and size of microcavity. For Fabry-Perot (F-P) cavity, the mode volume is approximately equal to the volume of the cavity. For whispering-gallery mode cavities, the mode volume is approximately the product of the microcavity's circumference and the cross-sectional area (on the order of the square of the wavelength).

The finesse F is defined as the ratio of the free spectral range (FSR, the frequency difference between adjacent resonant peaks) to the mode FWHM

$$F = \frac{\text{FSR}}{\delta\gamma} = \frac{\tau}{T_r} \quad (4)$$

where T_r is the photon round-trip time in the cavity. The finesse establishes a connection between the intra-cavity optical power and the input optical power. When the optical field couples with the resonator, a portion of the energy enters the resonator while another portion transmits out of it. The intracavity energy (W_{cavity}) can be expressed as $W_{\text{cavity}} = P_d \tau$, where P_d is the intracavity optical power, which is equal to the difference between the input power P_{in} and the transmitted power P_{trans} . Consequently, the expression for the total power inside the cavity is given as $P_{\text{cavity}} = W_{\text{cavity}}/T_r = P_d \tau/T_r = P_d F$. Furthermore, the energy density u within the resonator is represented as

$$u = \frac{W_{\text{cavity}}}{V} = \frac{P_d \tau}{V} = \frac{Q}{V} \frac{P_d}{\omega_m} \quad (5)$$

From the equation, the ratio of quality factor to mode volume (Q/V) is proportional to the energy density, while energy density directly determines the strength of interaction between light and matter. Therefore, microcavities with high-quality factors and small mode volumes exhibit stronger light-matter interactions [26].

Sensing mechanism of optical microcavities

Mode shift, splitting, and broadening

Monitoring the transmission spectrum of resonant wavelength is a universal method to detect changes of environmental quantities

surrounding microcavities, such as the binding of analytes. Particles with a different refractive index is adsorbed on the cavity, which modifies the optical length and induces the external scattering, thus leading to mode shift, mode splitting, and mode broadening [27–29]. Mode shift can be measured through transmission spectrum (as shown in Fig. 1A) or reflection spectrum. The variation of mode shift is related to the polarization intensity of particles α_{ex} , the electric field intensity distribution E_0 , and the position of the particles r_i . The frequency shift $\Delta\omega_m$ at resonance ω_m caused by external particles can be expressed as Eq. 6.

$$\frac{\Delta\omega_m}{\omega_m} = \frac{-(\alpha_{ex}/\epsilon_0)|E_0(r_i)|^2}{2 \int \epsilon_r(r)|E_0(r)|^2 dV} \quad (6)$$

where $\epsilon_r(r)$ is the dielectric constant throughout the cavity [27]. Although the mode shift method is suitable for sensing of a wide range of physical quantities and analytes, it is susceptible to environmental instabilities like noise and vibration, impairing its practical use out of lab. Due to the bidirectional propagation allowed in whispering gallery mode (WGM) microcavities, clockwise and counterclockwise light can coexist, resulting in mode degeneracy. Particle absorption results in coupling between the 2 modes and the degeneracy is lifted, leading to a splitting [29]. The mode splitting frequency shift δ and the linewidth broadening Γ are shown in Eq. 7,

$$\begin{aligned} \delta &= \frac{-\epsilon_m \text{Re}[\Psi] f^2(r) \omega_m}{\epsilon_c V} \\ \Gamma &= \frac{-\epsilon_m^{5/2} |\alpha|^2 f^2(r) \omega_m^4}{6\pi c^3 \epsilon_c V} \end{aligned} \quad (7)$$

where $\Psi = 3V_s(\epsilon_p - \epsilon_m)/(\epsilon_p + 2\epsilon_m) \cdot f^2(r)$ is field spatial variation determined by the optical wavelength. ϵ_m , ϵ_p , and ϵ_c are external cavity permittivity, particle permittivity, and cavity permittivity, respectively. V_s is the size of the particle [24]. Compared to mode shift mechanism, mode splitting, as shown in Fig. 1B, is more accurate because of its immunity to the variation of ambience. With this scheme, detections of nanoparticles and biomacromolecule are demonstrated [30,31], showing outstanding detection limit and sensitivity. Although it has many advantages, mode splitting approach requires the quality factor of microcavities to be sufficiently high to enable the splitting differentiable. Absorption and scattering loss can be brought when particles are attached to microcavity surface, which arouses the broadening of resonance linewidth, as shown in Fig. 1C. Similar to mode splitting, mode broadening is also a self-referenced technique, unaffected by environmental perturbations. Compared with the former, it does not propose a high requirement for the quality factor, but requires the adhesion of analytes to induce loss to microcavities, and the resonator might degrade with time if adhesion is kept for a long time [32].

Lasing and gain enhanced sensing mechanism

Lasers play a crucial role in the field of optics due to their low threshold, narrow linewidth, and high light intensity. The development of optical microcavities has made laser devices more integrated and miniaturized, which excitingly enriches the biochemical sensing applications in live cell monitoring, biomolecular footprint tracking, and multicomponent gas sensing [8,31,33,34]. The operation of lasing and gain assistance further suppress the resonance linewidth through the compensation of intrinsic loss. With the decrement of spectral linewidth, the slight changes in resonance, such as mode shift and splitting,

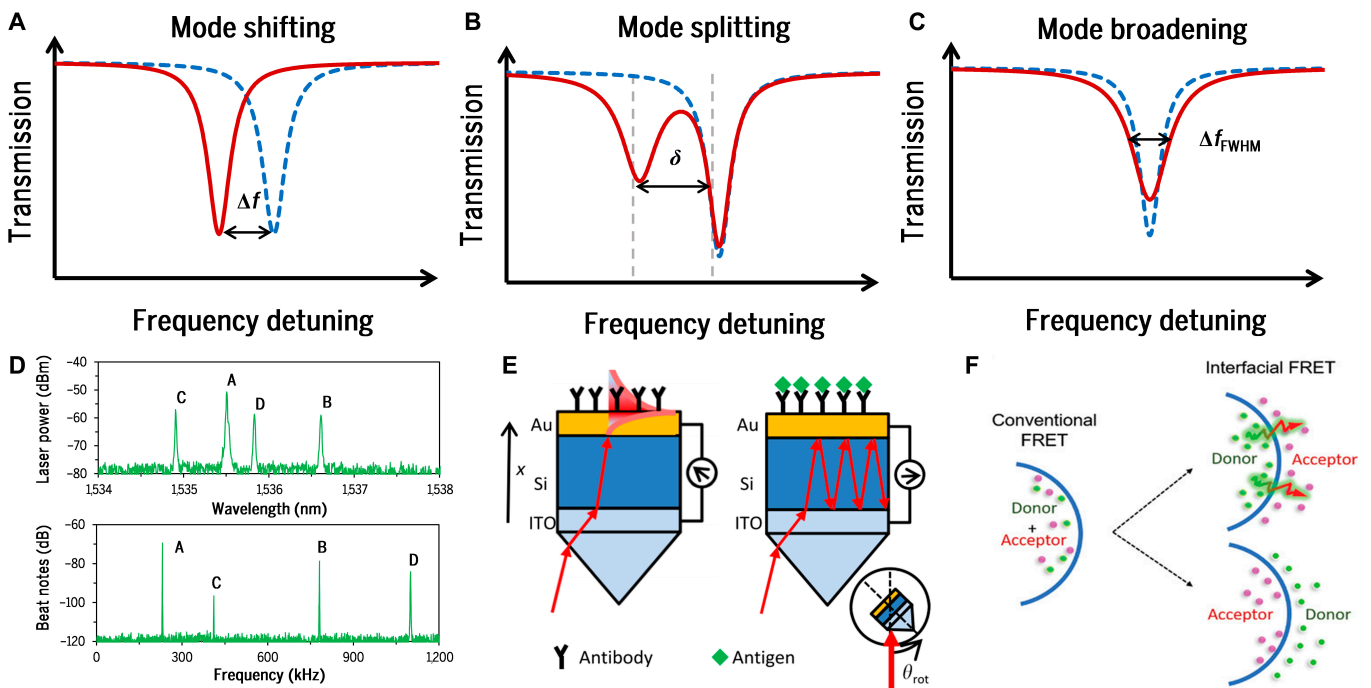


Fig. 1. Typical sensing mechanism for biochemical sensor. Mode shifting (A), mode splitting (B), and mode broadening (C). (D) Microcavity lasing for multi-component sensing mechanism. One microcavity can generate 4 laser modes (top panel) and 4 beat note signals (bottom panel), which can be regarded as 4 sensing channels for distinct targets [37]. (E) The mechanism of surface plasmon detection with Fabry-Pérot cavity [41]. (F) The mechanism of conventional FRET (left panel) and interfacial FRET (right panel) [51].

would be better resolved and clearly separated. Therefore, the sensing resolution can be greatly enhanced. Therefore, the detection limit can be improved remarkably [1,35,36]. In the experiment, the laser splitting mode could also be measured as beat note signals via the self-heterodyne detection technique, achieving both high-sensitivity and real-time detection [37]. Microcavities such as microsphere and microrod often support multiple intracavity transverse modes. These co-propagating modes offer a way to form multiple sensing channels for parallel biochemical detections. For instance, the top panel of Fig. 1D demonstrates that multiple optical lasers can simultaneously be generated in a single microcavity, from distinct mode families. When mode splitting occurs, these lasers can generate multiple beat note signals in a photodetector [37]. In spectrum, each signal can show independent sensitivities to varied targets; thus, they can be regarded as different sensing channels. For another example, in the lasing process, versatile Q-switching modes, orthogonally polarized modes, and coupling modes can also be considered as individual channels [38,39]. By comprehensively measuring the responses in these channels, one can analyze multispecies in mixture through calculating a matrix. Besides, it is worth mentioning that with advantages such as low threshold and high brightness, the microlaser device inside the cells enables cell tracking and dynamic product analysis with pump excitation [33].

Surface plasmon resonance

SPR is the resonance coupling between electromagnetic waves and the oscillation of charge density at the interface between a medium and a metal. The strong localized field generated near the surface by SPR greatly enhances the light-matter interaction [40]. As shown in Fig. 1E, researches have been dedicated to developing ultrasensitive sensors with the capability of detecting local tiny changes in the refractive index [41]. Combining SPR with microcavities offers several advantages. On one hand, it can enhance the overall optical response sensitivity of the hybrid system. Due to the coupling between the SPR mode and the microcavity mode, even small changes in the external environment, such as molecular adsorption or refractive index variations, can modulate the cavity mode [42]. On the other hand, due to the radiation and resistive losses in metal nanostructures, the linewidth of localized surface plasmon (LSP) resonance is greatly broader than that of waveguide modes supported by low-loss dielectric microcavities. Therefore, coupling LSP with microcavities is commonly used to reduce the resonance linewidth and improve the spectral resolution of detection [43]. In addition, SPR can contribute to surface-enhanced Raman scattering (SERS) spectroscopy. SERS provides a non-destructive and highly sensitive approach for fingerprinting and label-free detection of various molecules. SERS is mainly caused by the enhancement of surface electromagnetic [44]. Electromagnetic enhancement typically involves 2 processes [45], local field enhancement and radiative enhancement. When exciting surface plasmons, the electric field intensity is greatly enhanced near the surface, resulting in local field enhancement. This leads to an increased Raman scattering intensity of molecules near the surface. When the Raman mode matches the resonance of the plasmon, the Raman scattering light undergoes a second enhancement, known as radiative enhancement.

Fluorescence resonance energy transfer

Fluorescence resonance energy transfer (FRET) is widely used in biochemical sensing due to its high sensitivity, good selectivity,

and simplicity. The basic principle of FRET involves 2 different fluorescent molecules, a donor and an acceptor, as shown in Fig. 1F. The donor molecule absorbs excitation light and transitions from its ground state to an excited state. In its vicinity, there is an acceptor molecule. Energy is transferred from the donor to the acceptor through dipole-dipole interactions, causing the acceptor to transition to an excited state and emit fluorescence [46,47]. FRET is a non-radiative energy transfer process, eliminating the need for photon involvement and simplifying the sensing process. The emission spectrum of the donor should overlap with the absorption spectrum of the acceptor, and the typical distance between the donor and acceptor is less than 10 nm [48]. The optical microcavity with compact size, low threshold, and high emission efficiency serves as an excellent platform for FRET. Based on the characteristic of microcavity and the distance-sensitive nature of FRET, the binding of biomolecules to the donor-acceptor pair can alter the distance between them, thereby affecting the efficiency of energy transfer. Sensing of biomolecules can be achieved by observing changes in the output wavelength spectrum and energy transfer efficiency [49]. Biomolecules as part of the donor or acceptor, with the microcavity as a FRET platform, the concentration of the molecules could be measured by directly observing the intensity of the output wavelength [50]. Furthermore, it is a novel idea to separate the donor and acceptor molecules at the droplet microcavity interface. The excitation of acceptor molecules using induced light near the cavity interface through coherent radiation energy transfer has been proposed [51].

Optical frequency comb spectroscopy

As one of the most promising and powerful techniques, optical frequency comb (OFC) has a wide range of applications, such as ranging, coherent communications, and, especially, high-precision label-free sensing of biochemical molecules [16,52–55]. OFC consists of a series of equidistantly spaced comb lines with outstanding stability and broad spectrum. Therefore, it is an ideal broadband coherent light source for spectroscopy, which is used to interrogate molecular fingerprints with ultra-high resolution and fast acquisition speed. OFC spectroscopy can be classified into several schemes, such as direct frequency comb spectroscopy, Ramsey-comb spectroscopy, dispersive spectrometer-based spectroscopy, Michelson-based Fourier transform spectroscopy, and dual-comb spectroscopy [56–60]. Among them, dual-comb spectroscopy attracts the highest interest. Two combs with a slight repetition rate difference are generated, one goes through the analytes, and the other one acts as a local oscillator, beating with each other in the photodetector and transforming the spectrum into directly detectable radio frequency domain, as shown in Fig. 2A. This method is freed from geometry, and it is the only spectroscopy technique that can be applied for arbitrary span and spacing [61].

Optomechanical interaction

The optomechanical interaction in microcavities has been widely explored and applied in biochemical sensing, thanks to the narrow linewidth, easy-observable megahertz signals, low power consumption, and the characteristic vibrational modes of molecules, which contribute to high-resolution detection and molecular barcode establishment. A simple physical model of opto-mechanism can be explained by F-P cavity, where one end has a fixed reflective surface, while the other end has a freely

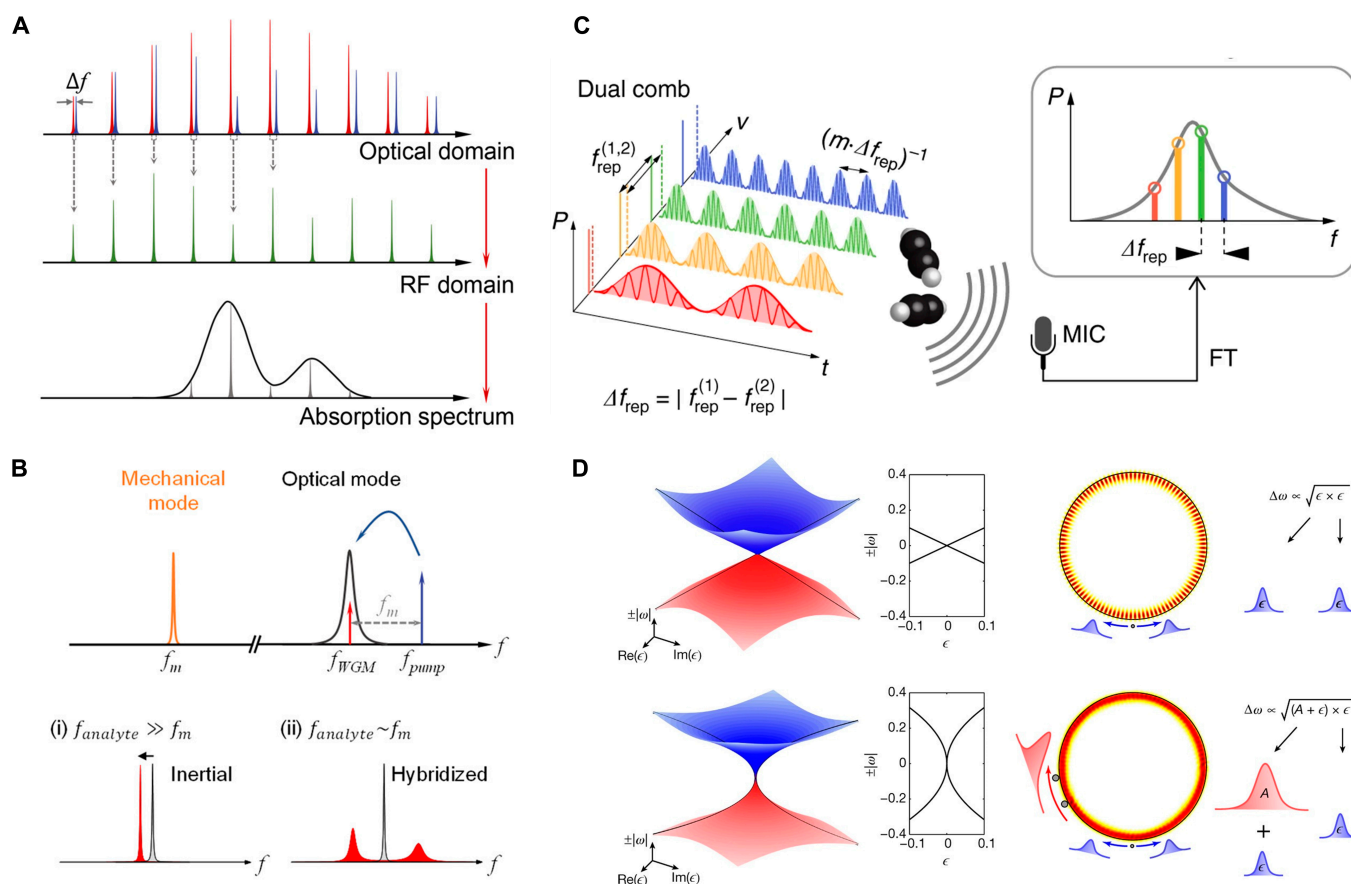


Fig. 2. Sensing enhancement mechanism for biochemical sensor. (A) Scheme of dual-comb spectroscopy. Two soliton comb lines (top panel) with slightly different repetition frequencies, radio frequency comb (middle panel) with repetition frequency and absorption spectrum (bottom panel) [65]. (B) Theory of optomechanical interaction in microcavities (top panel) and the different analyte–sensor interaction regimes (bottom panel) [65]. (C) Dual-frequency comb photo-acoustic spectroscopy uses broadband dual-frequency combs, where signal is composed of multiple heterodyne acoustic tones [69]. (D) WGM nanoparticle sensing mechanism of traditional diabolic point (top panel) and exceptional point (bottom panel) [36].

movable reflective surface. When light is injected, the cavity generates both mechanical oscillation modes and optical modes due to the radiation field. These 2 modes interact with each other. The radiation field changes the resonance frequency and damping rate of the mechanical mode, while the mechanical vibration changes the cavity length and the frequency of the optical mode [62]. Optomechanical systems consisting of small-sized, low-power-consumption, and on-chip integrated cavities can greatly enhance the interaction between optical and mechanical modes, providing new mechanisms for the detection of biomolecules and electrically neutral particles [63]. During the sensing process, when biomolecules, viruses, proteins, etc., adsorb to the surface of the cavity, the change in mass leads to a variation in the mechanical mode of the cavity, enabling sensing. Furthermore, strong interaction occurs when the adsorbed molecules have mechanical vibration modes similar to the mechanical mode of the cavity, resulting in marked changes in the cavity's mechanical mode and achieving sensing [64,65], as shown in Fig. 2B. In addition, the acoustic phonons generated during Brillouin scattering induce structural mechanical oscillation, and sensing is realized by detecting changes in the Brillouin mechanical mode.

Photothermal and photoacoustic spectroscopy

The utilization of photothermal effect and the emergence of highly sensitive acoustic detectors have brought attention to the

photothermal and photoacoustic spectroscopy method for biochemical sensing. When analytes absorb photons with the frequency of ν , parts of molecules undergo a transition from the ground state E_0 to the excited state E_1 . The energy absorbed through collisions between molecules in the excited state and the ground state is converted into translational kinetic energy, i.e., heat, through a non-radiative relaxation process. When periodically modulating the pump frequency, the analyte temperature exhibits periodic variation, leading to periodically changed pressure [66], as shown in Fig. 2C. Thermal-induced refractive index modulation can be probed by the techniques of diffraction and interferometry, and the analyte concentration can be analyzed by the amplitude variation of the acoustic signal. For microcavities, the change of material refractive index resulting from material absorption of light can be incisively captured by the shift of resonance, which demonstrates individual nanoparticle detection [43]. The aforementioned methods can detect slight energy absorption, even in cases of high reflectance and weak absorption, which work as a candidate for compact gas and medical sensors with excellent sensitivity, selectivity, and stability. Experiments have been designed to achieve individual nanoparticle detection and gas sensing [67–69]. However, realizing low-cost, high-performance light sources and ultrasensitive acoustic detectors still poses challenges for miniaturized and integrated photoacoustic sensors.

Exceptional point

Exceptional points (EPs) are a degeneracy of open non-Hermitian physical systems, for example, parity-time (P-T) symmetric system, where lots of intriguing phenomena have been observed, such as directional lasing, coherent perfect absorption, and sensitivity enhancement. In particular, the combination of singularities with optical microcavities holds significant potential for a wide range of applications [10,70–73]. Compared to the traditional Hermitian system with diabolic points (DPs), the EPs in open dissipative non-Hermitian systems are more sensitive to small perturbations, and the detection of individual nanoparticles has already been achieved [10]. This characteristic can be described using the Hamiltonian matrix, which represents the system energy. For a second-order coupled system, the eigenvalues in a Hermitian system are orthogonal to each other, and when subjected to perturbations, the change in eigenvalues $\Delta\omega$ is proportional to the perturbation strength ε . In contrast, the eigenvalues in a non-Hermitian system are not orthogonal, and when subjected to perturbations, the change in eigenvalues $\Delta\omega$ is proportional to the square root of the perturbation strength, as shown in Fig. 2D. Furthermore, as the order N of the EP increases, the sensitivity to perturbations follows $\Delta\omega \propto \varepsilon^{1/N}$ [72,74]. This implies that when constructing a non-Hermitian system and placing it at an EP, the system responds strongly to external perturbations, such as the adsorption of nanoparticles or biomolecules. Moreover, the smaller the perturbation is, the system's response will be more pronounced. However, due to the presence of quantum noise, the mode splitting caused by perturbations needs to exceed half of the noise and the linewidth of the split modes to be observable by instruments. To address this issue, Yang and colleagues introduced optical gain by doping erbium particles to compensate for the losses caused by scattering inside the cavity while maintaining the EPs. With increasing gain, the originally invisible mode splitting gradually becomes clear, thereby improving the sensing sensitivity and resolution and effectively avoiding detection errors [36]. Additionally, although the principle of EP sensitivity is independent of the cavity's material, geometry, intrinsic losses, or gain, when constructing a P-T symmetric system, such as a P-T conserving system, it often requires a gain cavity and a loss cavity. The loss is inherent to the cavity itself, while the gain is provided by gain media.

Structures, Materials, and Fabrications of Optical Microcavity-Based Biochemical Sensors

The structure and material of optical microcavities determine light-matter interactions, as well as application situation of the devices. Here, this review primarily introduces structures including F-P microcavities, WGM microcavities, photonic crystal (PhC) microcavities, and on-chip microcavities. Correspondingly, this review presents the material platforms and functional materials that support and extend these structures. Furthermore, this review showcases current mainstream approaches for microcavity fabrication and material functionalization.

Structures of optical microcavities

The F-P structure and WGM structure are fundamental resonant structures, with miniature size reaching micrometer and nanometer scales. Moreover, with the impact of nanoscale defect engineering, PhC microcavities present abundant flexible design possibilities. It is also worth mentioning that, with

the complementary metal-oxide-semiconductor (CMOS) process, integrated photonics and on-chip microcavity devices have shown revolutionary processes.

The F-P structure was first proposed by Charles Fabry and Alfred Perot. The F-P cavity consists of 2 parallel mirrors. The light coupled from ends through optical fibers or prisms propagates back and forth within the cavity, forming standing waves and evolving into stable modes. F-P microcavity has demonstrated important potential applications in nonlinear optics, optical filters, and highly sensitive detection [75–79]. Particularly in the realm of biosensing, the F-P optical cavity holds unique advantages and an indispensable position. On one hand, it possesses open-access geometry where the optical mode distributes in small refractive index space instead of dielectric, allowing direct interaction with biochemical molecules, as shown in Fig. 3A [3], which enhances sensing sensitivity. On the other hand, it exhibits high optical field intensity and strong interference effects. Minute changing signals of intensity, frequency, and phase caused by variations in effective refractive index are precisely captured, which confers a distinctive advantage in small-molecule biosensing [80]. In practical applications, to enhance the interaction between the evanescent field within the fiber core and external molecules, D-shape fibers, microfibers, and hollow fibers are designed and widely employed for F-P microcavities [39]. Furthermore, with the development of microfabrication processes, modifications and designs at the fiber end faces have become possible [81] and cascaded F-P cavities with Vernier effect have been proposed, as shown in Fig. 3B [82,83]. In addition, F-P cavities are combined with PhC and WGM microcavities to increase sensitivity and expand application scenarios [79,84].

WGM can be traced back to 1910, Lord Rayleigh figured out the physical essence behind the acoustic wave propagation in the dome of St Paul's Cathedral [85]. Later, it was proved that the same theory can be applied to electromagnetic wave as well, and intense researches have been dedicated to investigating the WGM microcavities. WGM microcavity confines light in a circular ring boundary through total internal reflection [86]. Due to its wavelength selectivity, the light that satisfies the following equation can be self-reinforced while others are eliminated, $m\lambda = 2\pi nR$, where m is integer, λ is resonant wavelength, n is refractive index, and R is cavity radius. The light in WGM microcavity is coupled externally. Near-field evanescent coupling scheme is often adopted for WGM microresonators, which requires that the evanescent field should be overlapped with intracavity modes, and phase-matching condition should be met [87]. The first scheme is prism coupling, by which the evanescent field is excited when the laser is incident on the prism. This method is easy to handle, with relatively high coupling efficiency, but the prism is cumbersome, hampering its integration. Slant-cut optical fiber can also be employed for WGM coupling. The meticulously cut and polished fiber is used to couple the light into the cavity. This way is more miniature, but fabrication of slant-cut fiber is complicated and time-consuming [88]. Another coupling approach relying on tapered fiber emerged. Stretching the fiber, the middle tapered section would be very narrow, leading to the presence of evanescent field around this region [89]. This method improves coupling efficiency up to 99%, while the drawback is that the tapered fiber is fragile and hard to encapsulate. Thanks to the flourishing development of microcavity manufacture technique, various types of WGM microcavity have emerged, including

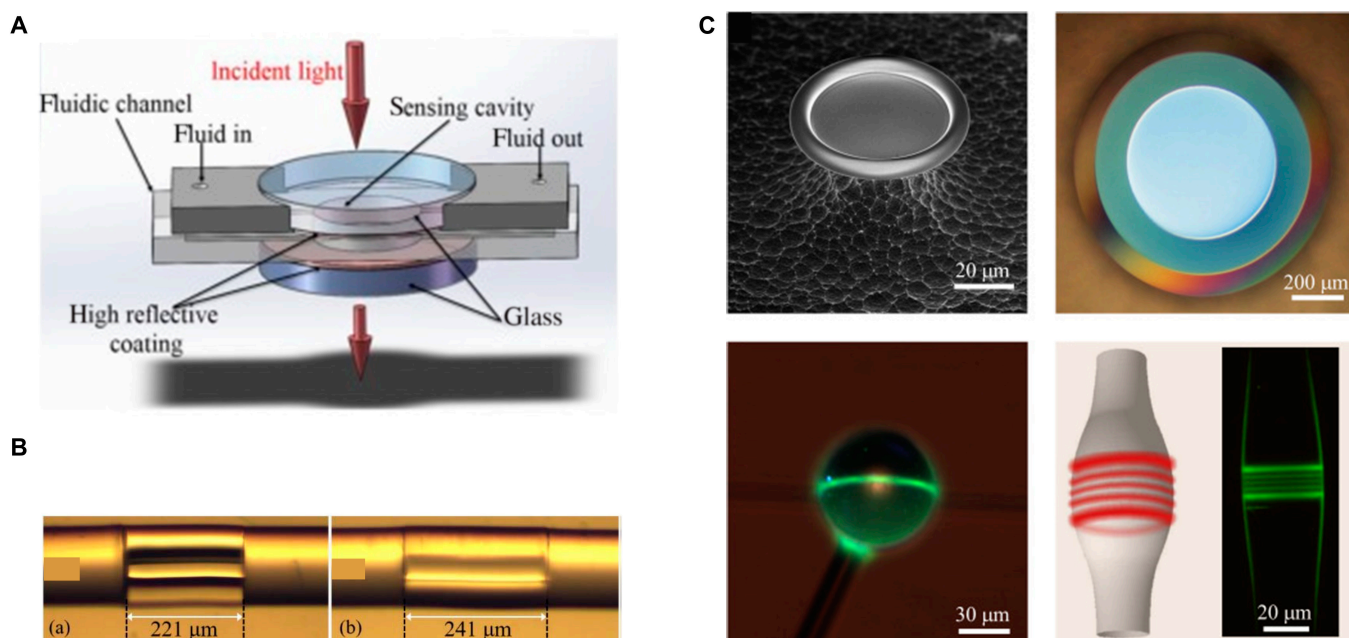


Fig. 3. Basic structure of microcavities. (A) Open Fabry-Perot (F-P) structure [200]. (B) Cascaded F-P microcavities [82]. (C) Typical whispering gallery mode microcavities, microtoroid, microrisk, microsphere, and microbottle [1].

microsphere, microtoroid, microdisk, microbottle, microbubble, and so on [90–94], as shown in Fig. 3C. With atom-scale surface roughness [95], the lifetime of photon trapped in the cavity can be prolonged substantially, leading to high-quality factor (up to 10^9 or even higher) [96]. Due to the small footprint of WGM microcavities, a tiny mode volume is ensured, enabling large optical density inside [97]. Moreover, the evanescent wave existing at the cavity boundary is sensitive to environmental perturbation, while a photon would interact with analytes millions of times during its lifetime, dramatically enhancing light-matter interaction and making WGM microcavity an ideal platform for biochemical sensing [1].

PhC is an original structure commonly found in nature, which operates the diffraction and reflection of light and generates rich colors. It is skillfully engineered and composed of periodically arranged dielectrics materials with different refractive indexes [98]. The periodic structure results in some light being unable to pass through the PhC, similar to the concept of a bandgap in atomic systems. According to their periodic arrangement in different dimensions, PhCs can be classified into 1-dimensional (1D), 2-dimensional (2D), and 3-dimensional (3D) structures. 1D PhCs are also known as Bragg mirrors. Currently, 2D PhCs mainly include 2 forms, PhC fibers and slab-PhCs [99], as shown in Fig. 4A and B, respectively. The cladding and core of the PhC fiber are composed of outer-layer pores arranged uniformly along the axial direction and inner layer dielectric materials. With the property of photonic bandgap, light can propagate along the length of the optical fiber [100]. 3D PhCs have a lattice and are periodically arranged along 3 mutually perpendicular axes, with common examples being opals [23]. PhCs, especially PhC microcavities, are highly advantageous for biochemical molecule sensing. Leveraging the photonic bandgap characteristics and excellent optical field localization, the process of molecular adsorption can be sensitively observed in PhCs and manifested in changes in transmission spectrum. Additionally, the high-quality factor and

resonant structure of PhCs can effectively reduce the linewidth of detection signals, facilitating the detection of small perturbations and lowering detection limits.

Rapid development of silicon photonics over the last several decades has spurred various passive components with reduced footprint and low fabrication cost, one of which is the microring resonator, which emerged as the potential candidate for the planar platform of photonics in the future, due to its compactness and mature fabrication process. A microring resonator, as shown in Fig. 4C, confines light through a silicon core, which is deposited on a silicon oxide cladding and covered by a low index cladding on the top. The waveguide loops back to itself, and when the optical path length is equal to multiple times of some specific wavelength, the light nearby can be trapped in the cavity and resonance can be formed constructively. The dimension of a waveguide cross-section is often 400 to 500 nm in width and 200 to 250 nm in height [101]. The high contrast between the cladding and core enables microring resonator with strong light confinement, enabling substantial footprint reduction since larger bending can be permitted. Moreover, its dispersion can be engineered through designing the cross-section of waveguides; either anomalous [102] or normal dispersion [103] can be tailored, demonstrating the high plasticity of microrings. The coupling of microring relies on the evanescent field between the microring and adjacent bus waveguide. There are 2 commonly used configurations: the first one is all-pass, where the microring is only coupled with one bus waveguide; the second one is add-drop, where the microring is between 2 parallel bus waveguides. The advantage of the latter is that intracavity field can be directly measured through the second waveguide [104].

Materials of optical microcavities

The diverse range of materials provides a solid foundation for enhancing the performance and expanding the functionality of optical microcavities. Technological breakthroughs and

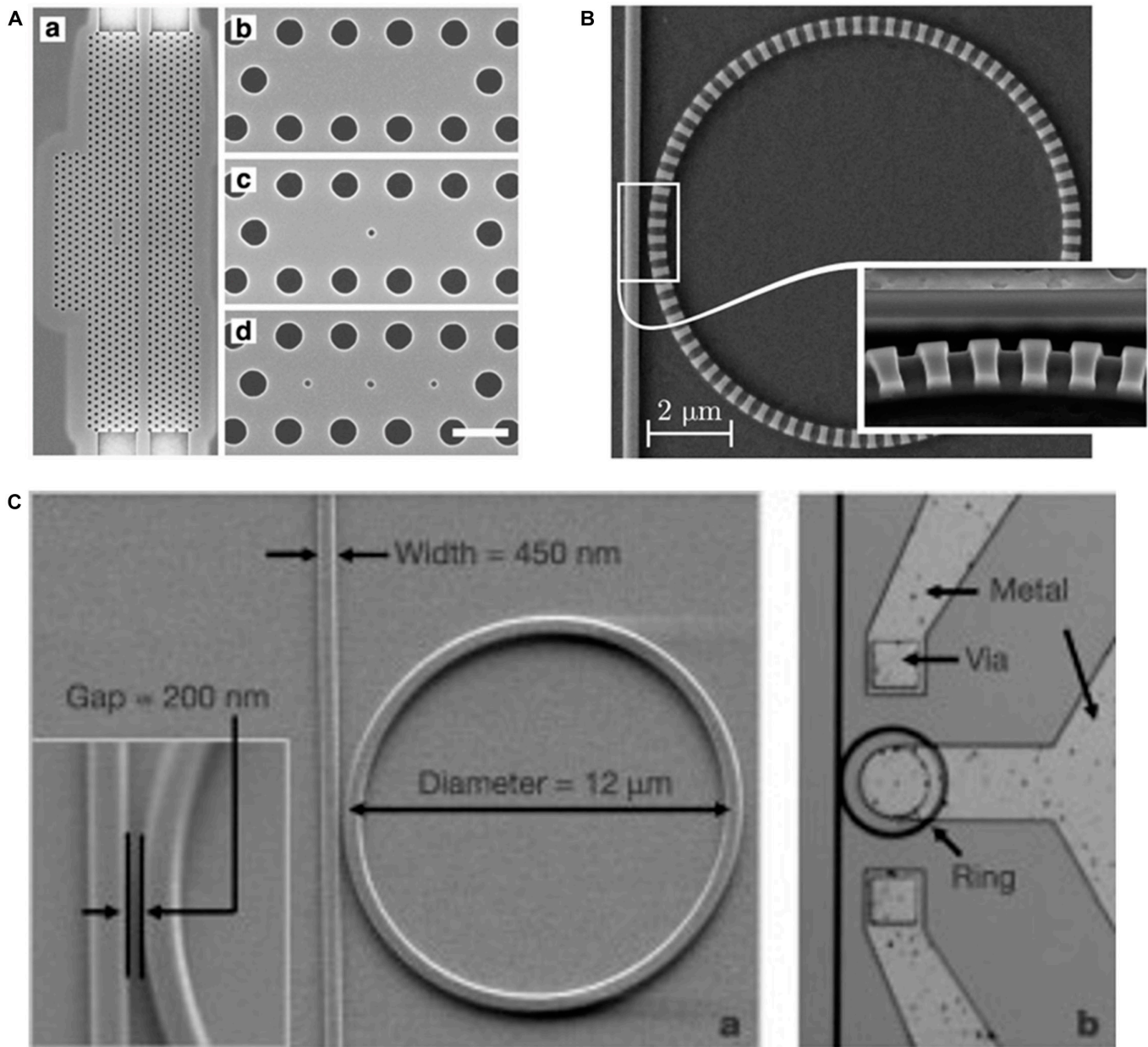


Fig. 4. Basic structure of microcavities. (A) Fabricated PhC microcavity (a), solid L3 cavity (b), single-hole defect cavity (c), and multiple-hole defect cavity (d) [201]. (B) Scanning electron microscopy image of the fabricated photonics crystal microring resonator [202]. (C) Scanning electron microscope image (a) and top-view microscope image (b) of the fabricated device [203].

principle explorations are closely related to the development of new material platforms. Therefore, this section primarily introduces the crucial materials that constitute optical microcavities and enable their functionalization, along with their unique properties.

Foundational materials platform for microcavity

The optical properties of materials, including their refractive index, bandgap, nonlinear coefficient, and quality factor, determine the material's optical behavior. The most common material is silicon dioxide (SiO_2), also known as silica, which is highly valued for its low absorption in the communication wavelength range, stability, and abundant availability. Silica has a refractive index of approximately 1.45, and due to its

symmetry, it possesses zero second-order nonlinear coefficient. Derived from silica, silicon (Si) is a typical semiconductor material, serving as a cornerstone of contemporary semiconductor technology. It is often doped with III–V elements to form n-type and p-type semiconductors. While silicon has undergone extensive researches and boasts mature processing techniques, the lack of even-order nonlinearities and the free carriers and 2-photon absorption in the C-band tremendously restrict its application. As a ramification of silicon, silicon nitride (SiN) possesses a wide transparent window and substantial nonlinear coefficients, making it an extremely attractive microcavity platform [105]. The low optical loss and advanced processing techniques have enabled silicon nitride on-chip microcavities to possess Q factors as high as 10^7 . In addition, silicon carbide (SiC)

possesses a high refractive index (~ 2.6) and exhibits strong second-order and third-order optical nonlinearities. The grinding and polishing approach has enabled SiC microcavities to achieve Q factors beyond 10^6 [106]. Another material, aluminum nitride (AlN), has a wide transparent window from violet to infrared, exhibiting second-order nonlinearity and providing a broad platform for studying photon–photon and photon–phonon interactions [107]. Similarly, lithium niobate (LN) also possesses high second-order and third-order nonlinear coefficients, enabling it to achieve high conversion efficiency in processes such as sum-frequency generation, second-harmonic generation, and cascade second-order nonlinear processes [2]. The exceptional electro-optic properties and light confinement capabilities of LN have made LN on insulator (LNOI) an important integrated optics platform. Aluminum gallium arsenide (AlGaAs) possesses optical gain and prominent second-order and third-order nonlinearity [108], while avoiding 2-photon absorption in the C-band, making AlGaAs a promising material for nonlinear optics applications.

It is worth noting that optical gain materials can compensate for inherent losses and enhance light–matter interactions. The use of optical gain materials is closely related to microcavity's structure, quality, and fabrication methods. Optical gain can be categorized into 2 types: gains based on energy-level systems and stimulated emission, and those based on nonlinear gain. The first type involves doping materials such as rare earth elements (e.g., erbium and ytterbium), quantum dots, organic dyes (e.g., rhodamine and fluorescein), and organic semiconductors (e.g., small organic molecules and conjugated polymers) [109,110]. The second type utilizes materials' nonlinear effects, such as stimulated Raman scattering and stimulated Brillouin scattering, to achieve optical amplification in the gain region. Optical gain holds promising prospects for enhancing Q factor of microcavities and improving device performance [111].

Functional material combination with microcavities

Functionalization with metallic materials enriches the sensing mechanism of microcavities. Metal materials support local surface plasmon resonance (LSPR) when appropriately excited by visible and near-infrared light. These LSPRs have strong field confinement (at the nanoscale) and enhancement near the metal surface, forming hot spots with high energy density. On one hand, the combination of plasmonic nanostructures and optical microcavities ensure the small mode volume of plasmonic mode and strong near-field enhancement. On the other hand, the absorption loss of metal nanostructures is compensated by optical microcavities [112]. The combination of plasmonic nanostructures and optical microcavities lead to a good balance between confinement and absorption losses [113].

2D materials exhibit a range of physical properties due to their distinct electronic structures, spanning from insulators to narrow-gap semiconductors, topological insulators, semimetals, and metals. Their natural passivation without dangling bonds allows easy integration with photonic structures like fibers and chips. Most 2D materials also exhibit excellent electrically tunable physical properties, making them well-suited for functional optoelectronic information devices. Existing reviews have extensively elucidated the amalgamation of 2D materials with diverse optical structures to achieve various functionalities [20]. In sensing applications, 2D materials serve as both chemically sensitive mediums and optical response enhancers, effectively addressing the insensitivity of traditional

optical structural materials to biochemical sensing molecules [114]. Graphene, the pioneering 2D material, holds a prominent place in the realm of 2D materials and finds widespread applications. With a substantial specific surface area and low electrical noise, graphene offers unique advantages [115]. When adsorbing polar molecules, the conductivity and refractive index of graphene change due to charge transfer. Notably, even at gas concentrations as low as parts per billion (ppb) or lower, graphene demonstrates a marked response [39]. Graphene also exhibits favorable adsorption and desorption characteristics. Two-dimensional TMDs, another class of materials, are widely employed due to their electronic properties that vary with the number of layers, tunability, weak van der Waals forces, and robust covalent bonds. In sensor applications, the sensing mechanism primarily relies on charge transfer and the adsorption of oxygen ions [116]. In comparison, black phosphorus is relatively less stable when juxtaposed with graphene and TMDs, and its sensitivity to the environment is particularly high. While often perceived as a drawback, this inherent sensitivity has transformed into an advantage in sensing applications [117].

Besides, antigens, antibodies, DNA, and RNA can be regarded as a special functionalization, because of the strong binding between antigens and antibodies, as well as the specific arrangement of deoxyribonucleotides and the principle of complementary base pairing [118,119]. These functional probes have become important tools in biochemical specific sensing, which effectively enhance sensing sensitivity and accurately identify target substances in complex environments.

Fabrications of optical microcavities

Based on the structures and materials, this section primarily introduces the fabrication methods of 4 typical microcavities and material functionalization processes. Understanding the fabrication process can boost design devices more effectively with better device performance.

Fabrication process of typical microcavities

Optical fiber F-P microcavities consist of reflective end faces or Bragg structures, resulting in a diverse range of mirror combinations. The emergence and development of femtosecond lasers have greatly enhanced fiber processing techniques and fiber fabrication methods. Techniques for cascaded F-P microcavities, as shown in Fig. 5A, writing FBG arrays, including weak and ultra-weak FBG arrays, are invented and high-quality factor is demonstrated. Besides, the 2-photon polymerization technique makes it possible to engrave the castle style F-P microcavities on fiber tip, where the periodically arranged square holes and intracavity polyvinyl alcohol enhance the humidity sensing greatly [120]. Furthermore, with the development of silicon-on-insulator (SOI) technology and integrated photonics platforms, on-chip F-P microcavities based on electron beam lithography (EBL) and reactive ion etching techniques, with an ultrahigh intrinsic Q factor of $\approx 3.4 \times 10^6$, have been proposed [79].

The WGM resonator contains various types of cavities, and distinct fabrication methods are employed for different structures. Microsphere is often fabricated with standard single mode fiber, and via electrical arc discharging thermal melting–shaping, surface tension pulls the melted fiber tip into a sphere [121] and gain element could be added in the melting process, as shown in Fig. 5B. The arc melting–shaping scheme ensures surface uniformity and smoothness, enabling ultrahigh-quality

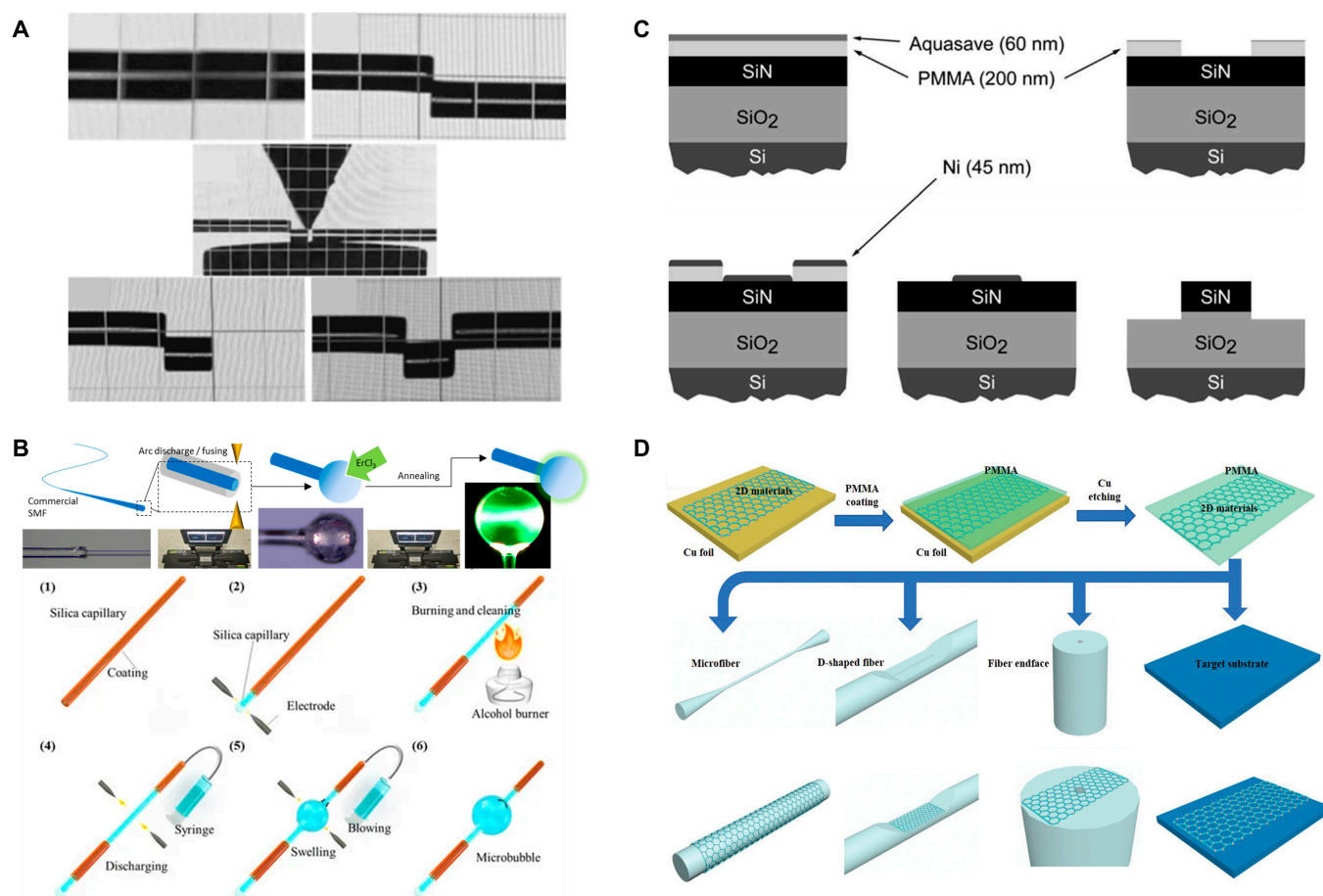


Fig. 5. Fabrication progress of microcavities and functionalization. (A) The fabrication process of on-line F-P microcavity microscopic images [204]. (B) Fabrication of the erbium-doped microsphere (top panel) [37] and microbubble resonators (bottom panel) [122]. (C) Fabrication flow of SiN microring [126]. (D) Schematic diagram of the wet transfer technology [20].

factor surpassing 10^8 . By controlling arc discharge power, discharge duration, and position, microspheres with different diameters can be obtained, which is usually hundreds of micrometers [16]. First demonstrated by Armani et al. [95], the fabrication flow of microtoroid can be divided into 4 steps, which are photolithography, pattern transfer, etch of silicon, and selective reflow of silica. The radius of toroid usually varies from 80 to 120 μm , and its Q factor surpasses 10^8 . The microdisk resonator can be fabricated conventionally with SU-8 on Si wafer, by using standard ultraviolet photolithography followed by several micro-fabrication steps such as soft bake and post-exposure bake [93]. Both microdisk and its waveguide can be produced on chip, excluding fragile components like tapered fiber and endowing it with high integrity. Microbubble can be obtained through the fuse-and-blow method, and its fabrication can be divided into several steps: sealing, melting, and air pressing [122]. The thickness of the wall and diameter can be controlled by the blowing time. In addition, hollow microbubble can usually be fabricated by different methods, such as fuse-and-blow (like mentioned above for microbubble), soften-and compress, and reduction of diameter with CO₂ laser [123].

The fabrication methods of PhC microcavities are varied from the dimensions, while 1D structure preparation is relatively simple, mainly including the immersion method, spray method, and spin-coating method. For 2D and 3D PhC microcavities, the mainstream fabrication methods could be summarized into

self-assembly and lithography techniques [124]. Bottom-up self-assembly methods, commonly used in colloids onto surface, do not require expensive and large-scale equipment and form the order structures more readily. The self-assembly methods mainly include spin-coating and interface methods, which rely on centrifugal force and surface tension, respectively [23]. With the development of the SOI technique, top-down lithography methods have turned into an important approach for PhC microcavities fabrication, owing to the precise fabrication of desired patterns and stable performance of PhC microcavities. EBL, nanoimprint lithography, electrochemical etching, and thin-film deposition methods are the major branches of lithography methods [125]. As shown in Fig. 5C, on-chip microcavities can be fabricated through Si wafer oxidation, Si₃N₄ deposition by CVD technique, poly(methyl methacrylate) and Aquasave layers placed on the top, scanning e-beam lithography, and Ni evaporation [126].

Fabrication process of functionalization

The assembly methods of metal nanostructures and 2D materials with optical microcavities have been summarized in the previous review [127]. For metal nanostructures, generally speaking, the functionalization and solution assembly with WGM cavity can be divided into 2 categories. One is to connect metal plasmonic nanoparticles and optical microcavities through non-covalent interaction, such as electrostatic bonding

or layer-by-layer polymer-assisted assembly. Another method is to functionalize the surface of the microcavity and then connect the metal plasmonic nanoparticles to the microcavity through covalent interactions, such as through surface silylation reactions. The emergence of 2D materials provides feasible solutions for improving the performance of traditional microcavity optical devices. Integrating 2D materials with microcavities will greatly enhance the light-matter interaction. The manufacturing methods of 2D materials mainly include mechanical exfoliation and liquid exfoliation, chemical vapor deposition, molecular beam epitaxy, wet chemical synthesis, etc. Among them, CVD and mechanical peeling are the most commonly used for manufacturing 2D materials. CVD is currently the most mainstream method for preparing large-area 2D materials [128]. Mechanical exfoliation is a method that utilizes the mechanical force generated by stretching adhesive tape to weaken the van der Waals forces between layers of 2D materials without disrupting the covalent bonds within the material's plane. This technique is a non-destructive preparation method that does not involve chemical reagents or reactions. Therefore, the resulting layered 2D nanosheets obtained by this method exhibit higher crystalline quality and purity, making them ideal candidates for investigating intrinsic material properties and optoelectronic performance. The transfer methods mainly consist of wet transfer and dry transfer. The wet transfer includes 3 steps, as shown in Fig. 5D. Firstly, spin-coat polymer on a substrate containing 2D materials to create a supporting and protective layer. Then, dissolve the substrate in a specific solution and use the target substrate to retrieve the 2D materials supported by the polymer. Finally, dissolve the supporting layer of polymer, rinse, and dry to complete the process. The dry transfer process involves 2 steps. Firstly, mechanically exfoliated 2D material is placed onto polydimethylsiloxane (PDMS). Then, align the 2D material with the target position and lower it until it makes contact with the target substrate. After heating, the original substrate and PDMS are lifted. In order to further improve the sensitivity and selectivity of sensors, some functionalized 2D materials and 2D material modification techniques have emerged. The most common functionalized 2D materials are reduced graphene oxide (rGO) and graphene oxide (GO). The longest used modification materials include metals, metal oxides, carbon nanotubes, and polymers [129–131].

The Progress on Optical Microcavity-Based Biochemical Sensor Devices

As mentioned earlier, the plentiful physical mechanisms with microcavity-enhancement and diverse designs combined with the optional functional materials have opened up broad opportunities for sensing and tracking in various physical states, dynamic and complexing conditions, multiple components, and selective biochemical processes. In order to showcase recent research advancements more effectively, this section will focus on specific sensing targets, divided into biological and chemical domain, which are both critical and typical in the application and research for environmental monitoring, clinical diagnostics, and industrial production. Therefore, in detail, biomacromolecules, cells and virus, solid particles, liquid ions, and gaseous molecules are elaborately introduced and discussed.

Biomacromolecules

Diverse protein molecules are common targets in biological detection. When protein molecules are used as acceptors in

FRET, energy transfer occurs and generates new frequencies when the donor substance is excited by pumping and binds to the protein molecules [8,22,132]. In 2020, Chen Yu-Cheng's research group proposed the concept of coherent radiative energy transfer by separating the donor and acceptor on the surface of a liquid crystal WGM microdroplet with high-concentration dye Coumarin 6 (C6) and the light-harvesting antenna phenomena. The detection limit was 20×10^{-6} M, and the laser threshold was on the order of $10 \mu\text{J mm}^{-2}$. Furthermore, based on the different absorption spectra of the adsorbed receptor molecules (Rhodamine 6G, Rhodamine B isothiocyanate, Rhodamine B, and Texas Red) on the droplet surface, the authors proposed the concept of a lasing-encoded microsensor and applied it to the detection of RhBITC and R-phycoerythrin protein molecules [51]. Building upon this, the research group designed a liquid crystal WGM microsphere sensor, as shown in Fig. 6A, for detecting extracellular vesicles secreted by 3D multicellular cancer spheroids in a complex liquid environment. C6-doped and antibody-functionalized microspheres (donor) can freely flow based on intrinsic convection and surface Marangoni pressure in the microspheres. When the microspheres come into contact with Nile Red-labeled extracellular vesicles (acceptor), fluorescence resonance energy transfer occurs, enabling protein sensing of vesicles at a level of 10^4 particles/ μL based on spectral intensity analysis, as shown in Fig. 6B. Additionally, the authors analyzed the spectral response of biomarkers in different tumor spheroids/subtypes, as shown in Fig. 6C, and generated cell-specific secreted protein markers [49]. Meanwhile, leveraging the specificity of antigen-antibody binding has opened up new possibilities in protein detection. In 2020, Yang et al. introduced a $4.7\text{-}\mu\text{m}$ -thin-walled optofluidic laser immunosensor based on hollow-core optical fiber (HOF). Under the influence of polyethylene glycol, antigens and immunoglobulin G antibodies formed immune complexes, which caused scattering losses and consequently reduced laser output. The size and concentration of these immune complexes increased with rising antigen concentrations and reaction times. Mixing laser dye rhodamine B with different antigen concentrations, the results show the response time of 15 min and the detection limit of 11 nM. By meticulously controlling preparation parameters, HOFs could be mass-produced with high reproducibility, enabling one-time, wash-free immunoassays [133].

Moreover, the longitudinal extension properties of the HOF were exploited to propose a wash-out-free sequential bioconjugation protocol. Yang et al. generated biotin sites on the inner wall of the HOF through capillary action, followed by the sequential bioconjugation of SAV-Cy3 molecules. Through competitive conjugation for avidin detection, this device achieved a detection of avidin concentration with a detection limit of 9.5 pM, a detection time of 10 min, and a sample volume of 589 nL [134]. Hollow optical fibers, as typical microring resonators and microfluidic devices, hold significant promise in the field of biochemical sensing, such as enabling multi-channel or array sensing [135]. Through spatial multiplexing, these devices achieve multi-sample, high-throughput detection. Precise control of the hollow optical fiber's geometry allows for highly reproducible multi-channel laser output, providing a novel approach to multi-channel, integrated biomolecular sensing devices [136]. Gong et al. introduced the concept of distributed fiber optofluidic laser. The gain medium within the hollow optical fiber is Rhodamine 6G (R6G) in quinoline, and laser excitation is introduced from the side through pulsed pumping. Five

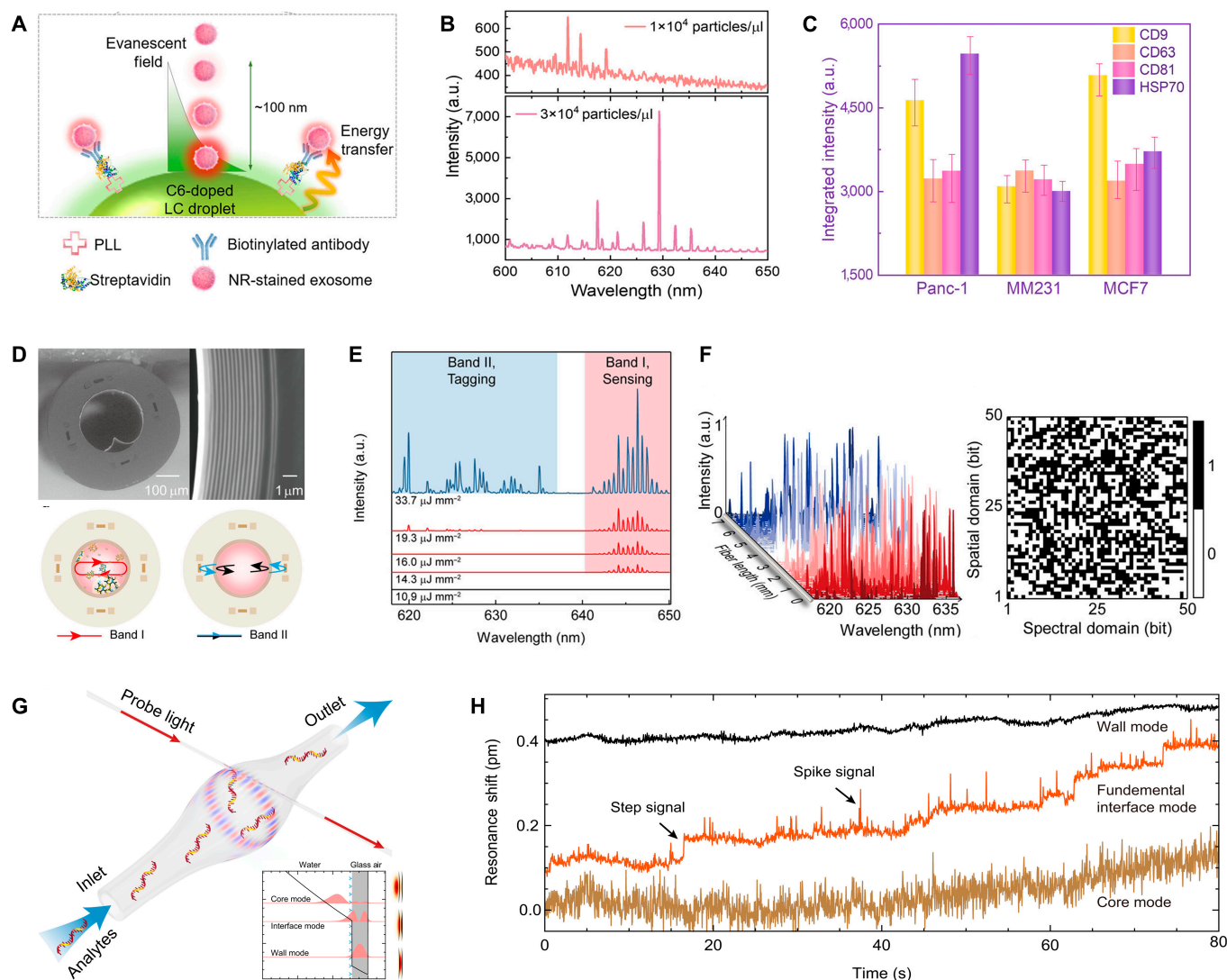


Fig. 6. The design and performance of biomacromolecules sensors. (A) The concept of interfacial energy transfer through the cavity interface from green laser emission to red laser emission [49]. (B) Lasing spectra of the self-propelled microlaser [49]. (C) Lasing spectral responses characterizing the abundance of protein biomarker expression [49]. (D) The scanning electron microscope images of the PBG fiber cross-section (top panel) and the schematics of the resonance of Band I and the resonance of Band II (bottom panel) [138]. (E) Emission spectra at different pump energy intensities, where the long-wave laser at low pump energy density for sensing (Band I) and the short-wave laser at high pump energy density tagging (Band II) [138]. (F) The spectra at different locations obtained by scanning along the fiber axis and generated laser tag [138]. (G) Schematic of the optofluidic microbubble biosensor, inset: effective potential of the microbubble cavity, radial field distributions, and the corresponding cross-sectional field distributions [140]. (H) Resonance shift induced by a single DNA oligo of a fundamental interface mode (orange curve), a wall mode (black curve), and a core mode (brown curve) with plasmonic enhancement [140].

parallel detection channels are longitudinally positioned along the hollow optical fiber axis. As the enzyme-catalyzed reaction proceeds, the light absorption of targets increases, resulting in a decrease in detected transmittance. As a result, the authors monitored the enzyme-catalyzed reaction catalyzed by horseradish peroxidase (HRP) and measured HRP concentration. Effective control of the distributed fiber optofluidic laser (DFOFL) wavelength over several hundred nanometers was also achieved by optimizing gain material concentrations and reconfiguring gain medium types [137]. Furthermore, as shown in Fig. 6D, Wang et al. utilized photonic bandgap (PBG) hollow-core fibers to achieve immunosensing of microalbumin (mAlb) and large-scale 2D laser coding. Two distinct resonant mechanisms exist within the gain-medium-injected PBG fiber, correspondingly generating the regions of periodic resonant peaks

(Band I) and random resonant peaks (Band II), as shown in Fig. 6E. The laser spectrum of Band I is used to detect light absorption and scattering caused by antigen-antibody complexes, with a detection limit of 0.06 ng/ μ l for mAlb. In the spectra of Band II, formed by longitudinally spaced PBG fiber spatial spectra, unique 2D laser tags for each fiber labeling are generated [138], as shown in Fig. 6F.

Functionalization of materials has also provided new approaches for highly sensitive protein sensing based on microcavities. In 2021, Allison et al. combined antibody-functionalized gold surface plasmon with a photovoltaic cell to construct an F-P structure, as shown in Fig. 1E. Light incident at a fixed angle excites electrons in the silicon layer, generating photocurrent. When the antigen is not adsorbed, the energy is mainly used for generating a small photocurrent from surface plasmons.

However, upon antigen adsorption, the change in surface refractive index reduces plasmon resonance, resulting in increased reflected light and photocurrent. By monitoring the photocurrent in real time, they achieved antigen (MBP-CoV-2NP) sensing with a sensitivity of 1.5 μA per micro refractive index unit (mRIU) and a response time on the order of minutes [41]. Wang et al. utilized WGM microsphere resonators and gold nanoparticle-functionalized DNazymes to detect L-histidine molecules with a detection limit of 5×10^{-16} (sub-femtomole) magnitude. Under the influence of L-histidine molecules, the gold nanoparticle-enhanced DNazymes are cleaved and bind to the receptor DNA on the inner wall of the microbubble cavity. This binding leads to excessive polarization of adhering biomolecules and changes in liquid crystal molecular orientation, resulting in spectral mode shifts of the WGM microcavity. This approach provides a new strategy for sensitivity enhancement in DNzyme-based sensing using gold functionalization [139].

DNA is a crucial genetic material found in living organisms. Sensing and detecting DNA have important implications for biodiversity research and disease prevention. Xu et al. designed an ultrasensitive WGM microfluidic sensor capable of detecting DNA with a molecular weight of 8 kDa in sample consumption as small as 10 μl , as shown in Fig. 6G. In high-Q microbubble cavity, WGMs were classified as wall mode, interface mode, and core mode, which could be distinguished by their distinct thermal responses. By optimizing the wall thickness of the microbubble cavity, the electromagnetic field was concentrated at the interface between the optical resonator and the analyte solution, achieving maximum sensitivity for interfacial molecular analysis. In experiments, the adsorption of biotinylated bovine serum albumin (BSA-biotin) molecules and streptavidin proteins caused a mode redshift, with a detection limit of 0.3 pg/cm^2 . The authors also introduced gold nanorods adsorbed on the inner surface of the microbubble cavity, enhancing the interface mode and enabling single-molecule DNA detection, as shown in Fig. 6H [140]. The principles of complementary base pairing endow DNA sensing with specificity and high sensitivity, driving the widespread application of DNA probes. In 2020, Zhang et al. mixed unlabeled DNA molecules and dye-doped liquid crystals in an F-P microcavity. They utilized the conformational changes of DNA as a switch to modulate the orientation of liquid crystals, resulting in laser emission switching between different wavelengths. This work sheds light on DNA sensing and bio-controlled lasers [141]. Zhang et al. proposed a DNA detection device based on a hollow-core fiber microbubble cavity. By attaching single-stranded DNA probes to the inner wall of the microbubble cavity with a thickness of 2.5 μm and a quality factor more than 10^6 , they achieved DNA detection of 10-mer and 25-mer length sequences by detecting spectral mode shifts. The detection volume was in the nanoliter range, with a detection limit in the picomolar range. The sensing sensitivity was 5.36 nm/RIU , and the resolution was 9.51×10^{-5} RIU. The device exhibited stable specificity and was capable of recognizing single-nucleotide mismatches in antigens [142].

Cell and virus

Cells are the fundamental units of living organisms, and further exploration of life requires the tracking of cell trajectories and investigation of cellular dynamics [67,143]. However, due to their micrometer-scale size, traditional optical microscopes are

unable to provide real-time and accurate cell positioning. In recent years, WGM microcavity-based fluorescence and laser emission, with their small volume (much smaller than that of cells), excellent biocompatibility, and low excitation threshold, have provided a promising avenue for cell sensing and tracking [144]. Himmelhaus and Francois [145] proposed a fluorescence emission sensing scheme by C6G-doped polystyrene WGM microspheres. The stress induced by live cells can lead to tiny deformations in the microsphere, which are reflected in the broadening and blue shift of the resonance, enabling the recognition of different stresses. By scanning the transmission spectrum, the sensor can detect real-time interactions from single cells. Furthermore, cells possess unique low-frequency vibrational information, which carry information on the structure and mechanical properties, enabling label-free sensing of biological information, as shown in the schema Fig. 7A. Gil-Santos et al. [31] achieved low-frequency vibration detection of single bacteria, *Staphylococcus epidermidis*, by closely matching the mechanical mode (546 MHz) of a microdisk to the bacterial vibration mode. Figure 7B presents the adsorption of the bacteria caused by substantial spectral changes. In particular, when the vibration frequency of microresonators is close to that of bacteria, their vibration modes hybridize, leading to marked changes in the spectrum. This work provides a powerful tool for the analysis of the vibrational spectra of individual biological entities. However, due to inherent low-frequency noise and cellular motion, low-frequency signals are often buried within the noise background, severely limiting the detection range. Jin et al. designed a CMOS-compatible heterodyne interferometer and proposed an up-conversion detection system. By introducing a signal offset, heterodyne interference amplifies low-frequency signals, making them detectable. When particles or bacteria are introduced into the sensing area, the beat frequency signal exhibits distinct, discrete steps. This device can detect a few attograms and single virus-like particles [146]. Furthermore, Tang et al. utilized high-quality factor WGM microsphere cavity to achieve real-time measurements of the natural vibrations of individual particles and the construction of single-cell vibration fingerprints. Particles and cells, when illuminated to pulsed laser irradiation, undergo natural vibrations that affect the intracavity WGMs. By analyzing temporal and spectral spectrum, the authors obtained biomechanical fingerprinting of microbial cells, which reflects cell size and internal structure. This offers an effective solution for label-free cell detection and targeted destruction of harmful cells [67].

From single-cell detection to cell tracking and the integration of cellular features, cell sensing research has progressed. In 2011, the group led by Seok Hyun Yun utilized genetically expressed green fluorescent protein within cells as an optical gain medium, forming a resonator with distributed Bragg reflectors on both sides. By using a 465 nm optical parametric oscillator as the light pump, the authors achieved lasing within cells and observed the characteristics of transverse and longitudinal modes [147]. Subsequently, the group proposed soft and hard WGM microresonators with high-quality factors, with “soft” and “hard” referring to the injection of natural lipid droplets or polystyrene microspheres into cells, respectively. Stress-induced changes within cells resulted in minute deformations of the microresonator shape, which were highly sensitive and reflected in the laser modes, enabling rapid labeling of numerous cells [148]. Furthermore, the group fabricated semiconductor

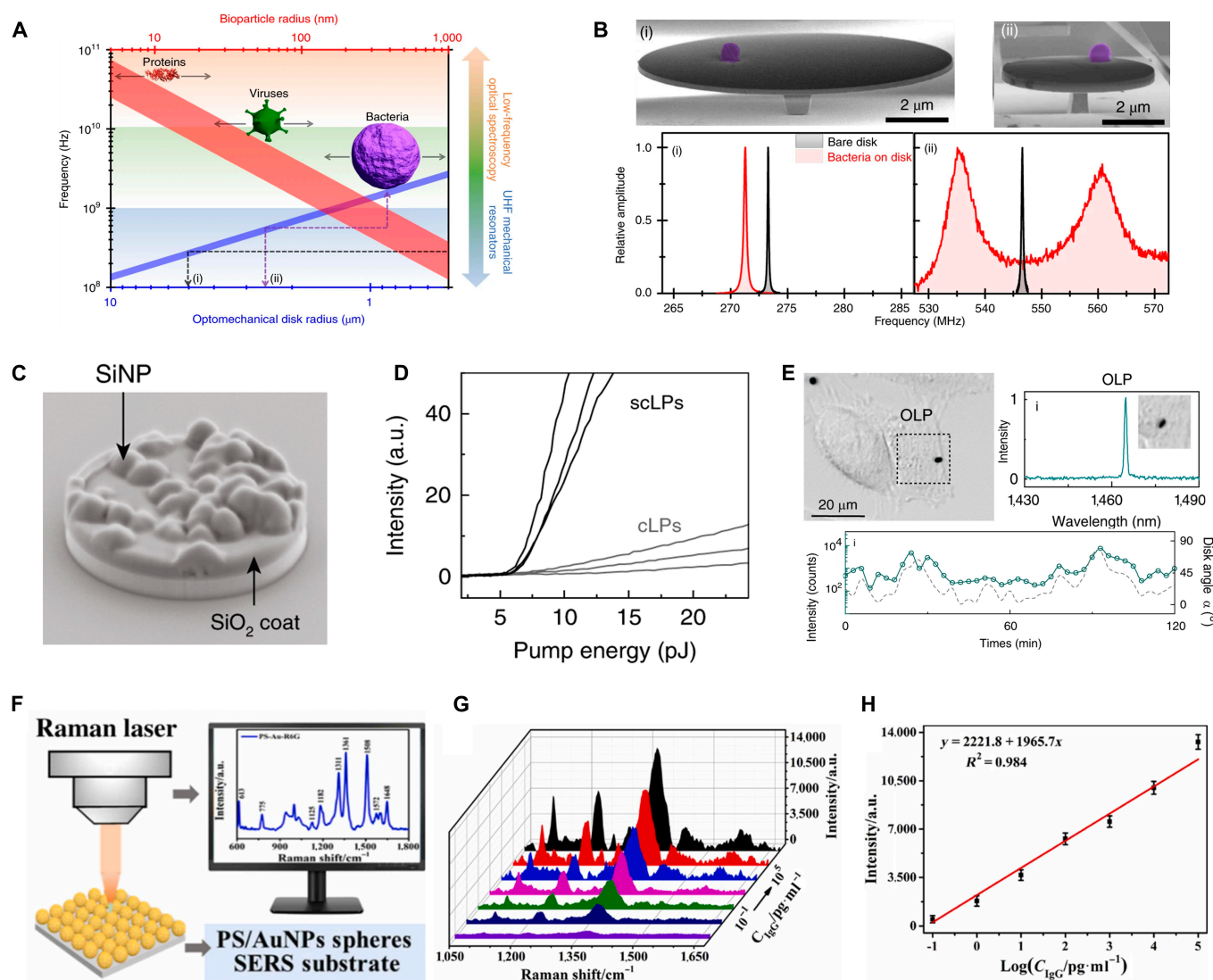


Fig. 7. The design and performance of cell and virus sensors. (A) Frequency of the radial breathing modes (RBMs) of a 320-nm-thick optomechanical disk (blue region) and of the fundamental mode of quasi-spherical biological particles adsorbed on a rigid support (red region) as a function of the disk and bioparticle radii [31]. (B) Scanning electron microscopy of 2 optomechanical disk resonators of radii 5 μm (i) and 2.5 μm (ii) with an attached *S. epidermidis* cell (top panel) and effect of bacterium adsorption on the RBM of the optomechanical disks (bottom panel) [31]. (C) SEM image of a laser particle with a scattering layer after detachment [33]. (D) Typical output curves versus pump energy for flat cLPs (gray curves) and scLPs (black curves) [33]. (E) OLP inside live HeLa cells and output spectra versus orientation (top). Lasing intensity (points) and orientation angle (dashed) traces as a function of time for OLPs [33]. (F) Sensing schemes of surface-enhanced Raman spectroscopy based on the microfluidic chip [158]. (G) Raman spectra with different concentration of Epstein-Barr virus. (H) Linear relationship between the SERS signal intensity at 1,361 cm^{-1} and the EBV IgG concentration [158].

microdisk cavities with silica coatings. These micro laser devices exhibited wide spectral range, sub-nanometer linewidths, low thresholds, single-mode excitation, excellent biocompatibility, and stability. Within a few days, the authors observed the behavior of thousands of individual cells within 3D tumor spheroids and obtained their behavioral characteristics in real time [149]. However, the aforementioned microlaser probes exhibited strong directionality when excited in the far field, and fluctuations in probe direction caused intensity variations during cell movement. To address this issue, as shown in Fig. 7C, the group introduced surface defects and scattering layers, achieving omnidirectional output of single-mode lasers with an FWHM of 0.25 nm. Figure 7D presents the high output intensity of a laser particle with a scattering layer and the threshold of 7 pJ. The slope efficiency, which characterizes the laser intensity change with angle, ranged from -24 to -6 dB for 100 microdisks.

High signal-to-noise ratio measurements were conducted continuously for 2 h in vitro with live cells, opening up new possibilities for more stable cell spectral identification and spatial tracking, as shown in Fig. 7E [33].

Microcavities with excellent biocompatibility, biodegradability, and low-threshold laser excitation have made cell dynamics tracking a hot research topic in the biological field. In 2018, Fikouras et al. fabricated semiconductor microdisk cavities with a diameter of 750 nm and quality factors ranging from 10^3 to 10^4 . The microlaser cavities, with their dimensions much smaller than cells, a low threshold of 0.13 pJ, and spectral stability with sub-50-ms jitter, achieved highly accurate dynamic monitoring of cells [150]. Subsequently, Schubert et al. injected fluorescent polystyrene microspheres with diameters of 10 to 20 μm into single cardiomyocytes. Under external pumping, the microspheres exhibited multimode laser emission. Contraction

of the cardiomyocytes resulted in temporal changes in the local refractive index of the microspheres. By observing the temporal changes in the spectra, the authors demonstrated that protein density within the myocardium is a potential cause of heart contraction [151]. To further explore the interactions of substances within single cells, Wang et al. proposed an active WGM microsphere cavity doped with DG (Dragon Green). The microcavity was introduced into cells (T47D, human breast cancer cell line) through co-culturing and endocytosis, enabling real-time measurement of R6G within the cells. This approach, based on spectral measurement rather than intensity, could be completed within a few seconds, effectively avoiding the influence of photobleaching and providing a viable solution for non-destructive sensing within single cells [152]. Furthermore, early detection of cancer cells provides crucial insights for timely intervention and clinical treatment. To better capture the dynamic processes of cancer cells and their products, structures with smaller volumes and higher luminescence intensity for labeling are currently under investigation and reporting. Guan et al. through the use of upconversion nanoparticles bound to small extracellular vesicles (sEVs), achieved super-resolution profiling that surpasses the optical diffraction limit [153]. Additionally, when combined with data analysis methods such as machine learning, cavity-based cancer early detection promises improved accuracy and sensitivity [154].

Viruses are generally made up of genetic material and proteins, as well as have a wide and profound impact on medical sciences and human society [155]. As a result, the detection of virus holds vital importance. The Vollmer group pioneered the sensing of individual viruses using WGM microspheres. They immersed a silica microsphere in a phosphate buffer saline (PBS) filled sample cell and injected influenza A (InfA) virions at a concentration of 10 fM. When polarizable nanoparticles attached to the surface of the microcavity, work was done by the evanescent field. Due to conservation of photon number in the cavity, the resonant photon frequency shifted [27]. Furthermore, Amarasinghe et al. functionalized antibodies for Ebola virus soluble glycoprotein (sGP) on a silicon photon microring resonator and achieved the detection of sGP. When the sGP solution flowed through the sensor's surface, sGP antigens were captured by antibodies, inducing a local refractive index change and resulting in a shift of the resonant wavelength [156]. SPR and spectroscopy also find extensive applications in virus detection. In 2021, Hao and Guo [157] integrated SPR into WGM optical microsensors, effectively enhancing the detection sensitivity of small respiratory viruses such as COVID-19. A biosensor based on a microfluidic chip by employing SERS was demonstrated, as shown in Fig. 7F [158]. The signal is amplified through synergetic coupling between the optical field of WGM microcavity and plasmon resonance of AuNPs (Fig. 7G), demonstrating that the detection limit of Epstein-Barr virus is $0.045 \text{ pg}\cdot\text{ml}^{-1}$ (Fig. 7H). Compared to traditional WGM sensors, SPR functionalization achieved a 3 to 5 times improvement in device sensitivity through rapid polymerase chain reaction (PCR) coupling, enabling fast, accurate, and specific virus detection. The sensor could detect individual airborne viruses as small as 20 nm. Direct frequency comb spectroscopy, utilizing a mid-infrared comb interacting with the gas, has also been employed in virus detection. In 2023, Ye et al. collected exhaled breath in a Tedlar bag and injected it into a gas chamber. By analyzing the absorption spectrum through machine learning, negatives and positives of SARS-CoV-2 infection could be differentiated, achieving real-time detection with a sensitivity reaching part-per-trillion volume [159].

Solid particle

Nanoparticle sensing with the size of tens to hundreds of nanometers is of great importance for atmospheric environment detection, and has also become an important research content in the field of biochemical sensing. In 2014, Ozdemir et al. used Raman gain of WGM microcavity and achieved the detection of NaCl nanoparticles of 10 nm radius. Inherent Raman gain compensates for the loss, since it enhances the Q factor and narrows resonance bandwidth. Moreover, Raman gain induces lasing in the splitting modes, and the distance can be identified through beat note in the RF domain, enabling detection and counting for a single nanoparticle [160]. Li et al. designed a silica microtoroid coupled with tapered fiber, as shown in Fig. 8A, and detected the 50 nm single nanoparticle in air or in an aqueous environment. In the sensing process, mode splitting is induced when the particle interacts with the microcavity. With lasers induced by Raman gain, beat note signal produced by lasers in the doublet is measured by the oscillation signal in the time domain as shown in Fig. 8B [64]. Heylman et al. detected single particles by exploiting on-chip microdisk cavities with the quality factor up to 10^7 . Optical absorption by the target leads to an increase in the microresonator's temperature, causing a resonance peak shift. This shift can be enhanced through lock-in amplification techniques to improve resonance shift sensitivities [43]. In addition to signal amplification strategies, reducing noise has emerged as another approach to enhance detection sensitivity. Zhi et al. demonstrated that attaching microfibers to microresonators can greatly reduce environmental-induced fluctuations, reducing the uncertainty of mechanical modes to 9.36 Hz within 5 min, resulting in a lower detection limit [161]. Han et al. proposed the structure of opto-mechanofluidic resonators, where flowing analyte solution is confined within the microchannel. Sensing is achieved through photon coupling to phonon modes, and the interaction between acoustic resonant cavity and particles is reflected by the resonance frequency shift [162]. In 2018, Jonathan et al. detected the single particle through microbubble resonators fabricated from silica capillaries, when nanoparticles are injected into the microbubble through a syringe pump. In the sensing process, nanoparticle interacts with WGM and leads to mode shift. At the same time, the dissipative effect of the nanoparticle would induce mode broadening. For the 100 nm particle size, the maximum shift is 400 MHz per particle, and maximum broadening is 100 MHz per particle [163].

In addition, EP sensing enhancement based on PT symmetry systems plays an important role in nanoparticle sensing [164,165]. In 2014, Wiersig combined the concept of EPs' enhanced sensitivity with microresonators. Through theoretical analysis and numerical simulations, they discussed the potential for single-particle detection [166]. Liu et al. [167] proposed that microresonators with P-T symmetric property, in comparison to loss-only microresonators, possess higher sensitivity, particularly at the in-phase transition point from unbroken to broken PT regimes. Subsequently, Chen et al. ingeniously designed experiments to validate the superiority of EP-based nanoparticle sensing. They achieved precise control of second-order EPs by tuning single WGM microring cavity with 2 nano-scatterers, as shown in Fig. 8C. When the target nanoparticle approached the microcavity, the transmission spectrum exhibited a splitting, as shown in Fig. 8D. In contrast to traditional methods, where the splitting of resonance is proportional to the perturbation intensity, the degree of this splitting was directly proportional

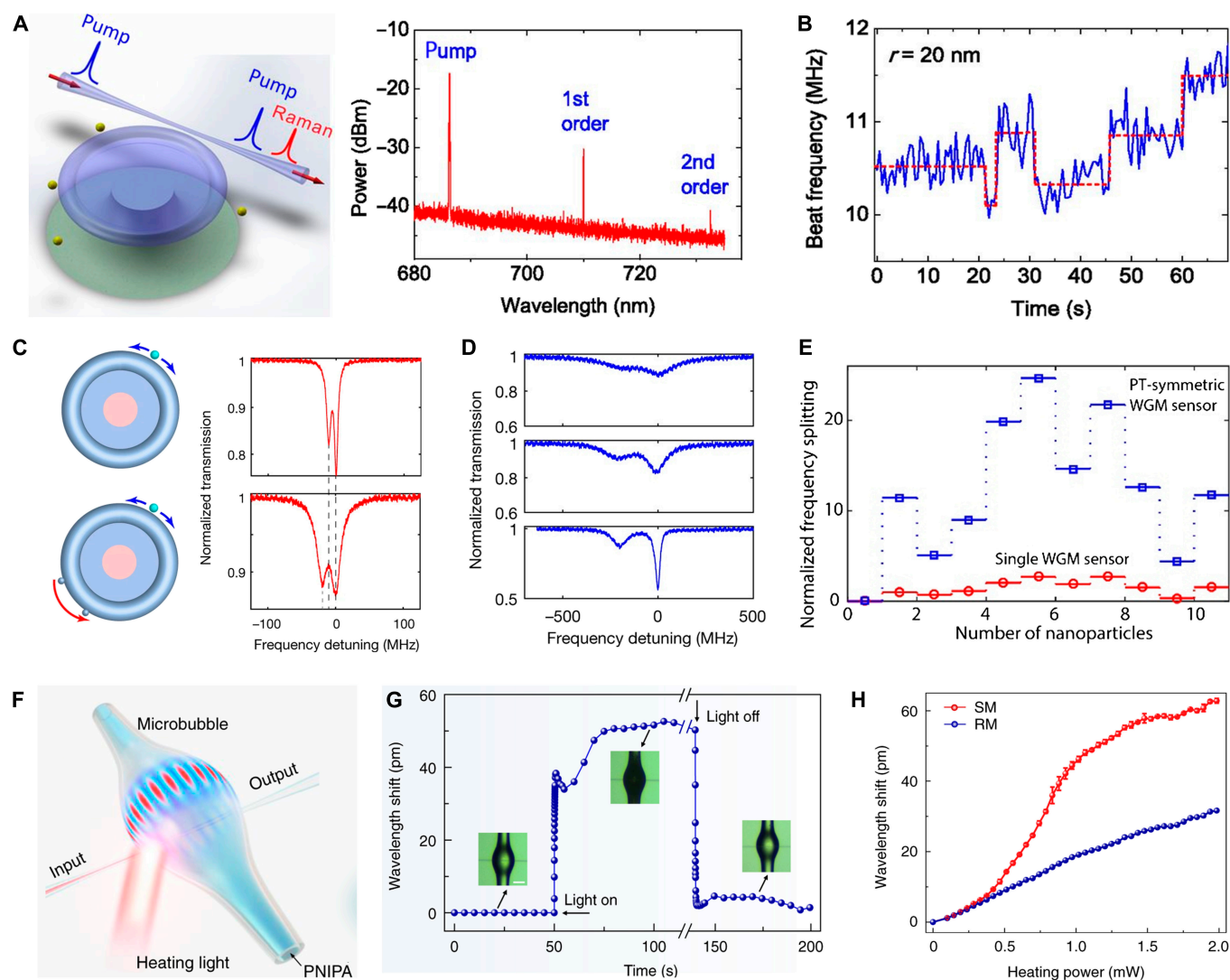


Fig. 8. The design and performance of nanoparticle sensors. (A) Schematic illustration of the taper-toroid coupling system (left panel) and the Raman spectrum of the microcavity in pure water with a pump power (right panel) [64]. (B) The beat frequency change vs. time when the chamber is filled with the solution containing PS nanoparticles with a radius of 20 nm [64]. (C) Adsorption of a target scatterer and the transmission spectra on the surface of the single cavity (top panel) and the exceptional-point sensor (bottom panel) [36]. (D) With the assistance of gain, the splitting modes that are initially barely seen (top panel) are clearly separated (bottom panel) [36]. (E) Numerical results of frequency splitting variation for 10 nanoparticles deposited on the PT-symmetric WGM sensor (blue lines and squares) and a single WGM sensor (red lines and circles) [168]. (F) Schematic of microbubble sensors for in operando monitoring of phase transition dynamics [169]. (G) Time-resolved wavelength shift of a typical WGM sensor during one cycle of heating-light on and off [169]. (H) Dependence of the wavelength shifts of reference mode and sensing mode on the heating light power [169].

to the square root of the particle perturbation intensity. Furthermore, the participation of erbium ion gain compensated for the cavity losses, enhancing spectral resolution and detection limits [36]. Building upon this, they sequentially detected the processes of multiple individual nanoparticles approaching one by one, further confirming the sensitivity enhancement due to EPs based on coupled resonators with balance gain and loss, as shown in Fig. 8E [168]. As sensitivity in detection continues to improve, researchers are increasingly focusing on the complex dynamics of particles under multiple effects. Yang et al., utilizing a self-referencing optofluidic microcavity (as shown in Fig. 8F), achieved dynamic detection of the phase transition of poly(N-isopropylacrylamide) molecules. The resonance mode of bubble microcavity experiences a continuous wavelength shift during one cyclic phase transition process with heating light on and off, as shown in

Fig. 8G. The material undergoes a phase transition upon exposure to heating light, affecting the reference mode and sensing mode within the microcavity, where Fig. 8H presents the dependence of the wavelength shifts of reference mode and sensing mode. By analyzing the transmission spectra of both modes, they decoupled and extracted information on refractive index and temperature changes during the molecular phase transition [169]. Furthermore, with advancements in micro- and nanofabrication techniques, novel structures distinct from traditional microresonators have been designed for particle detection. Tang et al. [170] introduced an on-chip dense waveguide sensor capable of detecting unlabeled nanoparticles as small as 100 nm. Yu et al. [171] leveraged the strong evanescent field of a nanofiber array, achieving a resolution of 10 nm for detecting individual 100-nm-diameter nanoparticles. This device is used for practical environmental

monitoring and was compared to officially released data, signifying its importance in monitoring ultrafine particulate matter in the air.

Liquid ions and organics

Sensing ions often require a solvent, and liquid sensing has become an important component of biosensing. Microcavities, especially microfluidic cavities, have been a hot research topic in liquid-phase sensing. Zhang et al. achieved reversible millimolar-level magnesium ion sensing in a microfluidic ring resonator using DNA Holliday junctions as mediators and employing the FRET technique, where the combination of DNA Holliday junctions was influenced by Mg^{2+} [172]. Motoki et al. proposed a dual microbubble cavity structure, where one microcavity was immersed in the test liquid for detection while another high-quality factor (1.65×10^7) cavity was placed in air for readout. By utilizing the mechanical mode coupling between the 2 microbubble cavities, damping and frequency shifts caused by changes in the viscosity of the surrounding medium or molecular adsorption in water could be read out through the optical modes in air, enabling high-sensitivity liquid-phase sensing in the environment [173]. Similarly, taking advantage of the highly sensitive characteristics of WGM microcavities, Samantha et al. achieved real-time detection of single droplets on a WGM microdisk, observing the dynamic process of droplet evaporation down to 20 aL. With a response time resolution below a millisecond, they studied the kinetics of rapid evaporation in a normal environment [174]. The liquid sensor based on photonics crystal microcavity has been reported. Kou et al. developed a pH colorimetric sensor based on a multilayer PhC, which exhibits a response time of 0.8 s. As the external pH changes from 4.0 to 8.5, the sensor's color shifts linearly from purple to red, increasing by 199 nm, and the stopband shifts by 7 nm per 0.1 pH [175].

Graphene and its derivatives possess exceptional physical and optical properties, such as a high surface-to-volume ratio, which have spurred liquid ion sensing. In 2016, Yao et al. deposited rGO onto a single-mode-multimode-single-mode fiber interferometer. By utilizing partially reduced graphene oxide (prGO) as a fluorescence quencher and enhancer layer, they achieved selective and highly sensitive sensing of dopamine, cadmium ions, and ssDNA. Specifically, fluorescence quenching occurred due to the adsorption of R6G by graphene. When external ions were mixed with R6G, the graphene could not adsorb R6G due to the adsorption of the ions, resulting in fluorescence recovery. In different pH environments, the adsorption capacity of GO for different ions varied, leading to different fluorescence recovery intensities and achieving selectivity. By measuring the spectral mode shift of the fiber interferometer, high-sensitivity sensing with detection limits in the nanomolar, micromolar, and picomolar range could be achieved for cadmium ions, dopamine, and ssDNA, respectively [176]. Similarly, Cao et al. deposited prGO on the inner wall of a capillary, and connected the ends to a single-mode fiber coated with gold to form an F-P cavity (Fig. 9A). By utilizing fluorescence quenching and fluorescence recovery, they achieved chemically selective measurements of dopamine, nicotine, and ssDNA, as shown in Fig. 9B. Besides, through lock-in amplification technology, the sensor achieved ultrahigh concentration detection and Fig. 9C presents a single-molecule dynamics process and the discrete jumps of the above molecules [177].

Plasmonic-based liquid biosensors have achieved high sensitivity and rapid detection in chemical sensing. Baaske and

Vollmer combined plasmonic gold nanorods with WGM microcavities, as shown in Fig. 9D and E, enabling single-ion sensing in aqueous medium and observing transient interactions between mercury ions and zinc ions with the nanorods [178]. Giaquinto et al. proposed a tunable optical cavity enhanced lab-on-fiber optrode incorporating smart materials as the active layer, as shown in Fig. 9F. The multi-responsive microgels sandwiched possess a swelling and collapsing mechanism, which induces the interaction of plasmonic resonance and cavity mode. Figure 9G shows the zoom-in wavelength shift for different molecule concentrations within 15 min and the results present the 7-nm shift for 160 μM . The optrode could achieve the detection of glucose in solution and act as a nano-optomechanical actuator triggered by light [179]. He et al. [180] achieved sensitive quantitative detection of Pb^{2+} by integrating chemical enhancement with electromagnetic enhancement using hybrid metal nanostructures and a single layer of graphene, demonstrating an enhancement factor of 3.85×10^8 . Chen et al. demonstrated the detection of p-nitroaniline using SERS with a detection limit of approximately 1.0×10^{-14} mol and an enhancement factor of 2.42×10^7 . These examples highlight the potential of SERS in achieving highly sensitive and selective detection in various applications. In addition, by employing techniques such as heterodyne detection and lock-in amplification, the influence of external noise and intermode interference was reduced, enabling single-molecule-level detection of the aforementioned molecules and providing a new pathway for high-performance sensor devices and systems [181]. Recently, Fakhri et al. [182] achieved concentration detection of NaCl, sucrose, and glycerol solutions by coating gold nanoparticles on unclad single-mode fibers, obtaining the highest sensitivity of 3,157.98 nm/RIU (glycerol solution) and a resolution of 3.16×10^{-5} (RIU) (glycerol solution). Furthermore, graphene-based field-effect transistor devices, which directly respond to external stimuli, have enabled high sensitivity and rapid response time in liquid detection [183–185].

Gas molecules

Gas molecule sensing plays a crucial role in environmental monitoring, industrial production, health diagnostics, and other fields. Compared to solid particles and liquid ions, gas molecules have smaller volumes and lighter masses, requiring higher sensitivity from gas sensors. Microcavities, with their high-quality factors and small mode volumes, exhibit strong light-matter interaction. One common approach to gas sensing is the detection of changes in the refractive index of gases. Ibrahim proposed a low-symmetry interferometric structure based on PhC, enabling label-free, all-optical gas and liquid detection. By monitoring the shift of Fano resonance peaks in the output spectrum, changes in the refractive index of the surrounding medium are sensitively captured. The sensitivities of gaseous and liquid analytes are 300 nm/RIU and 263.2 nm/RIU, respectively [186]. However, refractive index-based sensors may face challenges in discriminating between mixed gases. Due to the inert nature of the materials, microcavities often do not respond to gas molecules. 2D materials such as graphene, which serve as natural gas adsorbents and judges, can greatly enhance the sensitivity and enrich the sensing species. The integration of 2D materials with waveguides opens up a new pathway for stable, highly sensitive, and compact integrated gas sensors [80]. In 2014, Wu et al. deposited graphene on the side-polished surface of a multimode optical fiber,

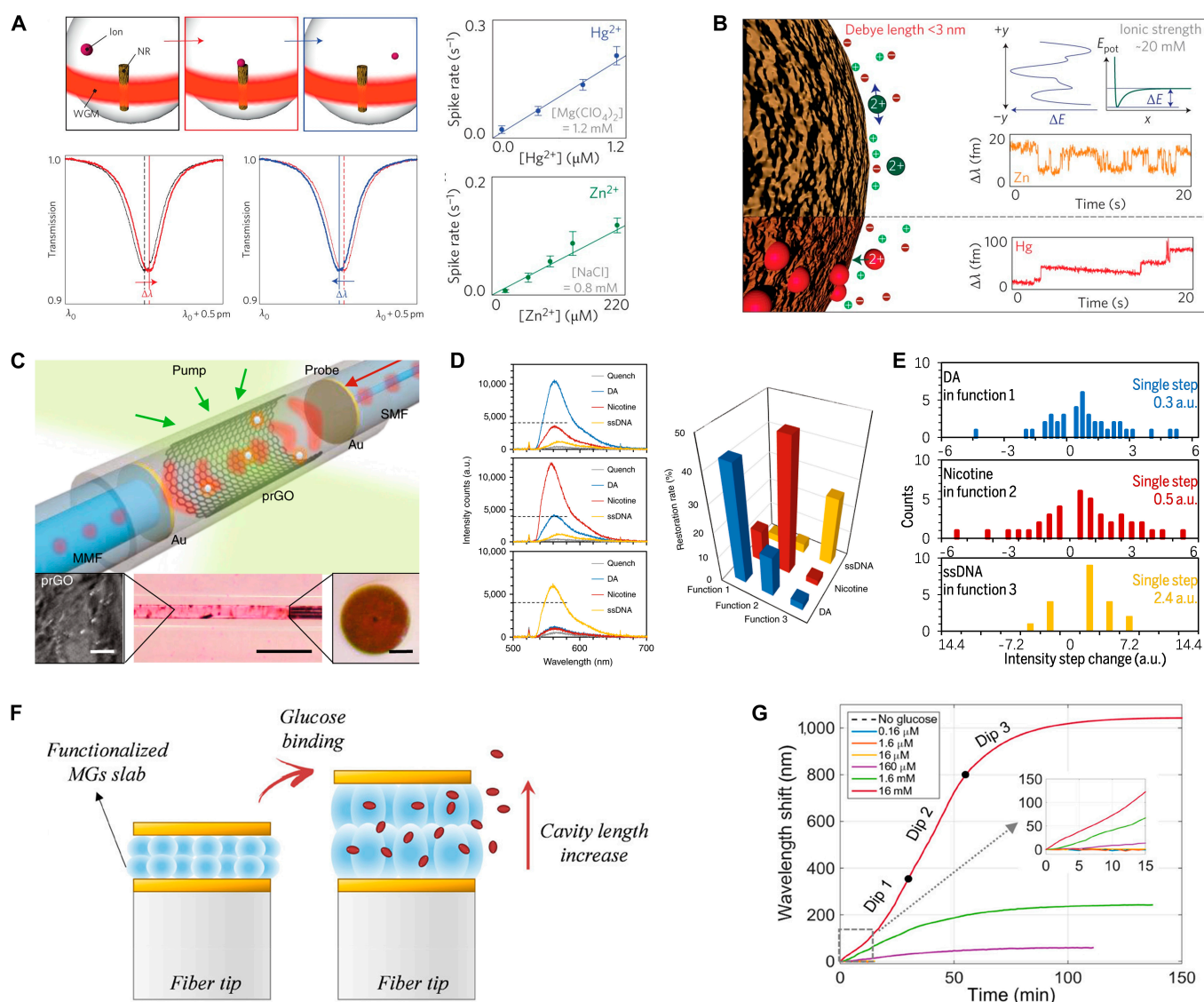


Fig. 9. The design and performance of liquid ions and organic sensors. (A) Schematic architecture of the device. Two reflectors are collimated in a silica capillary with the prGO deposited inside [177]. (B) Measured results for the sensor-target pairs [177]. (C) Statistical histograms show the discrete jumps in the fluorescence counts. After lock-in amplification, for DA, nicotine, and ssDNA, the standard deviations are 0.3, 0.5, and 2.4 mV, respectively [177]. (D) Transient interactions of single zinc and mercury ions with the NRs (left panel) and absolute values of these rates and concentrations (symbols) and corresponding linear fits (lines) for mercury and zinc ions (right panel) [178]. (E) At high ionic strengths, the Zn ions remain confined for longer in the local minima found along the NR's surface and the Hg ions bond permanently to the NRs. Step transitions toward the red and the blue side of the spectrum are observed for Zn, and toward the red for Hg [178]. (F) Schematic of the cavity device working in the etalon-like regime for glucose detection [179]. (G) Wavelength shift as a function of time for different glucose concentrations with the enhancement of cavity; inset shows a zoomed-in image of the first 15 min of the sensorgram [179].

forming a multimode fiber interferometer. By detecting spectral frequency shifts, they achieved sensing of ammonia and water vapor with sensitivities of 3 pm/ppm and 1 pm/ppm, respectively, and detection limits of 0.04 ppm and 0.1 ppm. Yao et al. deposited rGO on the inner surface of a high- Q (10^6) microbubble cavity integrated with microfluidics. Based on forward Brillouin light scattering, they realized the electron-photon-phonon interaction induced by gas adsorption. By detecting the spectral variations of the mechanical mode ($Q \approx 43,670$) beat frequency signal, they achieved a sensitivity of 200 kHz/ppm for ammonia molecules and a sensing limit of 1 ppb. Multicomponent sensing has been an important research direction in gas sensing [6].

Progresses have been made in highly sensitive and multi-component gas sensing by utilizing graphene-functionalized

composite microcavity structures and exploring various intrinsic physical mechanisms. Guo et al. designed a graphene-composite D-shaped distributed Bragg reflector structure. The structure consists of an erbium-doped fiber with fiber Bragg gratings (FBGs) at both ends, forming an F-P microresonator, and graphene transferred onto the polished surface of the D-shaped structure, as shown in Fig. 10A. This configuration exhibits low-threshold laser output and features 2 characteristic modes, laser-Q-switching mode and cavity oscillation mode, where Fig. 10B shows the beat note signals of them. During gas sensing, different gas adsorptions have distinct effects on these characteristic modes, resulting in different beat frequency spectrum shifts. By measuring the frequency shifts of the characteristic modes and utilizing the gas sensitivity coefficients,

concentration equations were established and solved, enabling multicomponent sensing of ammonia and carbon dioxide gases, as shown in Fig. 10C. The detection limits achieved were 1 nm/l in a vacuum environment and ppb level in ambient air [187]. Furthermore, Tan et al. deposited graphene asymmetrically on the surface of a high-Q microsphere cavity, enabling the observation of different families of Stokes solitons through the microsphere cavity, as shown in Fig. 10D. These Stokes solitons exhibited locked repetition frequencies and different central offsets. Through self-heterodyne detection, 3 sets of beat frequency signals (Fig. 10E) were obtained in the sub-gigahertz range. The adsorption of different gas molecules affected the transverse modes of the microsphere differently, resulting in distinct beat frequency shifts. By establishing and solving matrix equations, the group achieved the simultaneous

detection of ammonia, carbon dioxide, and water vapor at the picomolar per liter level. Moreover, through lock-in amplification techniques, the dynamic processes of single gas molecule adsorption and desorption were observed, reaching the detection limit of single molecules as shown in Fig. 10F [38]. Similarly, the graphene functionalized microsphere gas sensor was proposed, which achieved a detection limit of 4 ppb and a maximum sensitivity of 750 Hz/ppb for SO_2 gas. With the sensing mechanism of merge effect from sub-comb, this device not only resided in thermal-stable regime with high robustness, but also avoided complicated soliton procedures to allow plug-and-play operation, which properly combined microcavity comb and high-sensitivity detection [188]. Additionally, the microsphere cavity was transformed into a microlaser device by incorporating erbium-doped ions. Graphene deposition on

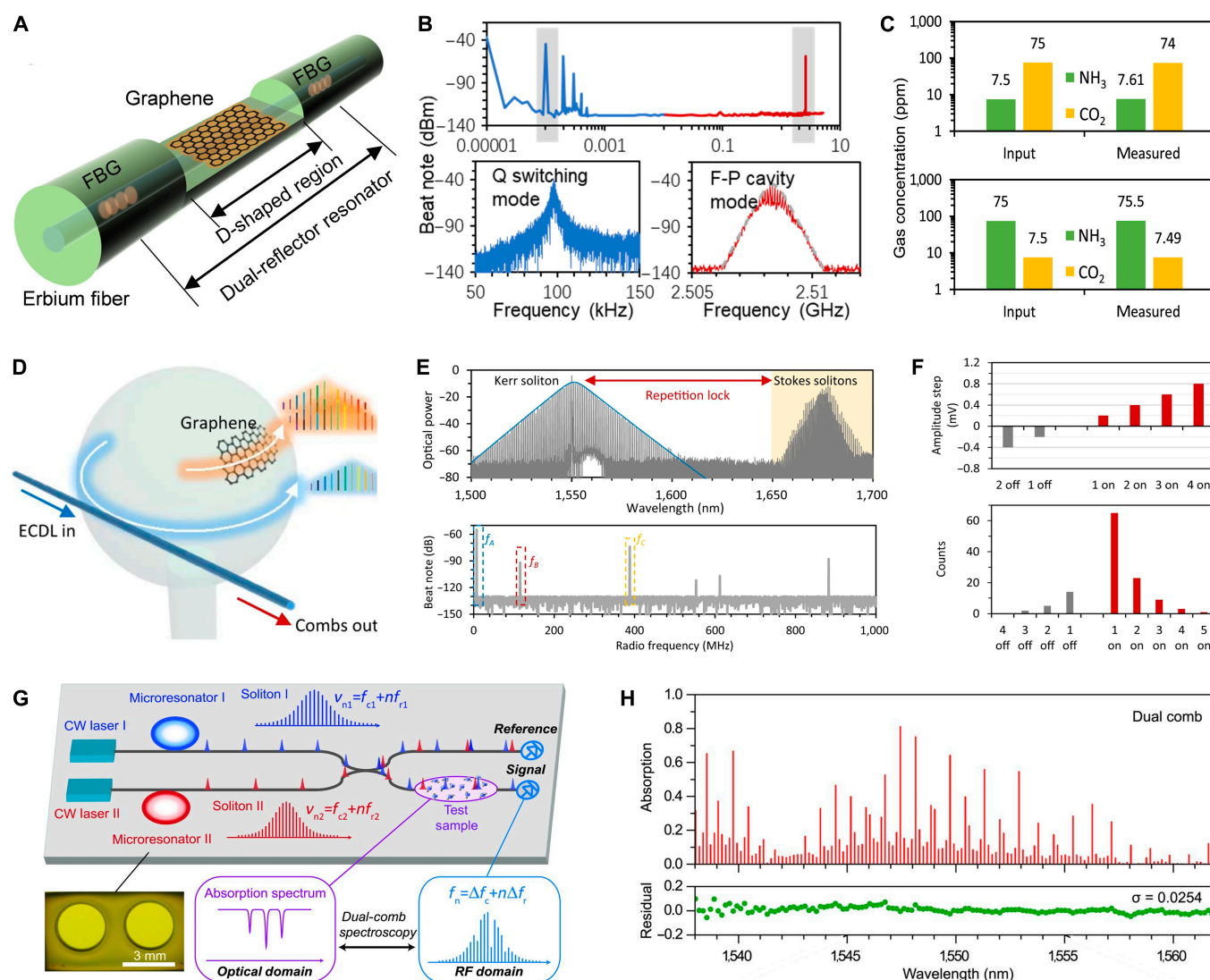


Fig. 10. The design and performance of gas molecule sensors. (A) Sketch of the structure. A graphene is deposited on the D-shaped Er-doped fiber region, and a pair of FBGs composes the microlaser cavity [38]. (B) Photoelectric heterodyne, the blue and red curve indicate the beat notes of Q switching and cavity repetition (top panel) and zoomed-in beat notes of Q switching and cavity repetition (bottom panel) [38]. (C) Gas identification for NH_3/CO_2 gas mixtures in dry air, with concentration ratio 1:10 (top panel) and 10:1 (bottom panel) [38]. (D) Schematic diagram of the device. An external cavity diode laser (ECDL) is used as the pump to excite soliton combs, which belong to distinct mode families [16]. (E) The measured optical spectrum shows both Kerr soliton and Stokes solitons (top panel) and beat note of the Stokes combs in the U band (bottom panel) [16]. (F) The measured steps are integer multiples of 0.2 mV (top panel) and the count of the molecular on/off (adsorption/desorption) events (bottom panel) [16]. (G) Microresonator-based dual-comb spectroscopy [190]. (H) Absorption spectrum band of H^{13}CN measured by direct power transmission using the microresonator-based dual-comb spectrum (top panel) and the residual difference between the 2 spectra is shown in green (bottom panel) [190].

the microsphere cavity surface induced backward scattering and laser mode splitting. By utilizing self-heterodyne detection, mode splitting in the sub-megahertz range was observed. The microsphere cavity supported multiple families of transverse mode patterns, resulting in the observation of multiple beat frequency signals in the hundreds of kilohertz range. By selectively choosing the gas-sensitive modes, real-time measurement of mode splitting variations enabled label-free, multicomponent sensing of ammonia, carbon dioxide, nitrogen dioxide, and water vapor at the ppb level. Furthermore, through lock-in amplification techniques, the dynamic processes of single molecules were detected [16]. Recently, An et al. integrated high-sensitivity Brillouin lasers with the aforementioned graphene microsphere structure. Thanks to the high-quality factor microcavity, this device achieved low phase noise of -161 dBc/Hz at 1 MHz. Under the influence of a pump laser, Brillouin lasers from different mode families were generated, and the beat frequency signals between these lasers were detected. Gas adsorption on the device resulted in shifts in the Brillouin laser beat notes, with different gases causing varying shifts in both direction and magnitude. Based on this, the device enabled high-sensitivity, all-optical multi-gas detection and the real-time monitoring of single-molecule dynamic processes for NH_3 , CO_2 , and NO_2 , with gas detection limits reaching the order of tens of parts per trillion and a mixing measurement error below 6% [189].

Dual-comb spectroscopy based on WGM microcavity is also an important method for high-sensitivity gas sensing. Two solitons are generated and stabilized in 2 on-chip microresonators with slightly different repetition rate by Vahala research group. One of them goes through a gas cell of molecules, while the other is used as reference. Then, absorption spectrum and reference comb are coupled together, as shown in Fig. 10G, and time waveform is recorded and transformed to frequency domain, which shifts the interfering soliton spectra to RF domain. Through this manner, the detection of H^{13}CN has been achieved, as shown in Fig. 10H [190]. Photoacoustic and photothermal techniques are important traditional methods for gas detection. Karhu et al. [191] demonstrated a simple light-emitting-diode-based photoacoustic NO_2 sensor with a sub-ppb detection limit. In 2022, Wang et al. obtained the photothermal spectrum of C_2H_2 pumped by dual comb source in as short as 10 ms by Fourier transformation of the interferogram generated by the F-P interferometer. This work achieved splicing of C_2H_2 spectra exceeding 1 THz, and achieved a minimum detection limit of 8.7 ppm under conditions of 15 mW dual comb power and 7 cm fiber length [192].

Conclusion and Outlook

Over the past decade, optical microcavity-based biochemical sensors have witnessed significant developments. These sensor devices, with their remarkable characteristics of high-quality factors, small mode volumes, and strong light-matter interactions, have proven to be versatile and powerful tools for detecting and quantifying changes induced by external substances. These developments have led to a wealth of biochemical sensing principles and related techniques, offering substantial potentials for diverse applications. Leveraging the advancements in materials and micro/nano fabrication schemes, optical microcavities have been crafted from a variety of material platforms with customized fabrication strategies, which are, therefore, tailorable to specific biochemical environments. The

blooming advances has brought substantial progress and groundbreaking achievements in critical performance metrics, including sensing sensitivity, detection limit, response time, and multicomponent selectivity. This review summarizes the performances of optical microcavity-based biosensor devices, encompassing applications that range from biomacromolecules, organic entities to solid particles, liquid ions, and gases, as presented in Table.

Nonetheless, it is important to note that many innovative principles and structural designs proposed in the current research are confined to specific contexts within the realm of biochemical sensing. The division between principles, structures, and device applications has created a barrier that limits the potential synergy between the sensitive enhancements introduced by principles and the excellent properties brought about by structural innovations. To overcome these limitations and unlock their full capabilities, future research endeavors may focus on bridging these gaps, fostering interdisciplinary collaboration, and exploring the untapped potential of these sensors across various biochemical contexts. In light of these insights, this review outlines possible research trends, challenges, and opportunities for the next generation of microcavity-based biosensor devices, to inspire further exploration in this field.

1. Exploration on active microcavities: Active microcavities present promising avenues for investigation. The introduction of gain through doping and nonlinearity can effectively counter microcavity losses [28]. The high- Q factor generates intense light fields, facilitating the excitation of higher-order modes, EPs [193], and nonlinearity [194], hereby enhancing the diversity and quality of sensing signals.

2. Interfacial mode modulation and interfacial binding amplification engineering: Understanding the role of microcavity interfacial modes in biochemical sensing sensitivity is crucial. External molecules primarily affect these interfaces. By modulating mode distribution, we can enhance the evanescent field at the interface [140]. The amplification of sensing signals is achieved by binding receptor molecules both inside and outside the interface [51].

3. Development of label-free molecular fingerprinting: The diverse modes and physical mechanisms of microcavities open doors to label-free molecular fingerprinting. This allows for the extraction of unique molecular vibrational frequency information [31,67] and mode barcodes [195]. These advancements establish multidimensional molecular fingerprint databases, enabling high-sensitivity label-free detection amidst complex background noise. Techniques such as multiplexing of optical spatial and frequency spectra, optical barcoding, and lasing encoding are widely applied in biochemical detection [144,196].

4. Multi-cavity combination, cascading, and arrays: Combining various microcavity types can lead to nested configurations, such as F-P combined with WGM microcavities [197]. Multi-cavity cascading [198], or multiple-signal in single-cavity configurations within a single cavity [16], enable ultrasensitive sensing analysis based on higher-order effects. Furthermore, microcavity arrays can be employed for multi-channel sensing [137,173].

5. Functionalization and optimization of on-chip microcavities: Overcoming the material inertness of on-chip microcavities to small biochemical molecules is essential. This enables their response to such molecules and leverages the advantages of on-chip microcavities for expanded biochemical functionalities [20].

Table. Recent advancements in optical microcavity-based biochemical sensors

Detection objects		Device structure	Quality factor	Sensing mechanism	Detection limit/sensitivity	Other performance	Ref.
Biomacromolecules	Photo-synthetic protein	Liquid crystal WGM microdroplet	–	Coherent radiative energy transfer by separating the donor and acceptor on the surface of microdroplet.	20×10^{-6} M	Laser threshold: $10 \mu\text{J mm}^{-1}$	[49]
	Immunoglobulin G antibody	Thin-walled optofluidic hollow-core optical fiber	1.67×10^6 in air	Immune complexes cause scattering losses and reduced laser output.	11 nM	Response time: 15 min; Lasing threshold: $1.67 \mu\text{J mm}^{-2}$	[132]
	Antigen (MBP-CoV-2NP)	F-P structure with antibody-functionalized gold surface plasmon	–	Surface refractive index changing reduces plasmon resonance and photogenerated current.	Sensitivity: $1.5 \mu\text{A/mRIU}$	Response time: several minutes	[35]
	DNA with a molecular weight of 8 kDa	WGM microfluidic microbubble cavity	2.6×10^8 (before) and 8.6×10^6 (after) gold nanorod adsorption	Interface mode enhances the sensitivity and detected by resonance shift.	0.3 pg/cm^2 for streptavidin proteins and single-molecule DNA	Sample consumption: 10 pl	[138]
Cell and virus	Single bacteria	WGM microdisk resonator	1,640 (mechanical Q)	Optomechanical interaction: the vibration modes hybridize and spectrum changes significantly.	Single bacteria	–	[31]
	Microbial cells	WGM microsphere cavity	$\sim 10^6$	Photoacoustics: cell's vibrational mode is stimulated photoacoustically by the absorption of laser pulses and acoustically coupled to optical resonance.	Single particle and single cell	Detection bandwidth: 1 GHz	[65]
	Cell tracking	Microdisk with surface defects and scattering layers	–	Omnidirectional output of single-mode lasers	Cell	Detection time in vitro with live cell: 2 h	[37]
Solid particle	Epstein-Barr virus	AuNP spheres	–	Surface-enhanced Raman spectroscopy	$0.045 \text{ pg}\cdot\text{ml}^{-1}$	–	[156]
	Dye-doped polystyrene spheres	Microbubble resonator	5×10^7	Mode shift and mode broadening	Single nanoparticle with radius of 100 nm	–	[161]
	Single particle	WGM microring cavity	More than 3×10^7 at 1,550 nm	Exceptional point enhancement: adjusting the position of scatters and keeping the system at EPs	Single polystyrene nanoparticle of radius 200 nm	–	[73]
	Atmospheric particulate	Nanowaveguide array structure	–	Particle-induced scattering	Single nanoparticle with radius of 10 nm	Real-time detection	[169]

(Continued)

Table. (Continued)

				Detection limit/sensitivity	Other performance	Ref.	
Detection objects	Device structure	Quality factor	Sensing mechanism				
Liquid ion	pH of the solution	Multilayer photonic crystal	–	Goal-directed molecular regulations by the post-hydrolysis strategy	Sensitivity: 7 nm/0.1 pH	Detection time: 0.8 s	[173]
	Nicotine, dopamine and ssDNA	Hollow-cavity Fabry-Perot (FP) resonator with partially reduced graphene oxide on the inner wall	–	Intracavity fluorescent resonance energy transfer	200 μM for DA and ~10 nM for nicotine and ssDNA	Selectal sensing	[175]
	Glucose	Tunable optical cavity enhanced lab-on fiber optrode	–	Microgels sandwiched induces the interaction of plasmonic resonance and cavity mode by swelling/collapsing mechanism.	16 μM	–	[177]
Gas molecule	NH ₃ and CO ₂ gas	F-P structure: Graphene-composite D-shaped distributed Bragg reflector	7.8×10^5	Gas adsorption affects Q-switching mode and cavity oscillation mode, leading to beat note signal shift.	NH ₃ : 2 nM/I (vacuum) and 10 ppb (air); CO ₂ : 6 nM/I (vacuum) and 36 ppb (air)	Multicomponent label-free sensing	[186]
	NH ₃ , CO ₂ , H ₂ O	Graphene functionalized over-modal microresonator	1.5×10^8	Absorption of gas molecule induces refractive index change of sensor, leading to variation of Stokes soliton offsets.	Single gas molecule	Multicomponents detection	[16]
	NH ₃ , CO ₂ , NO ₂ , and H ₂ O gas	Microsphere cavity functionalized with graphene	$\sim 5 \times 10^7$	Graphene induces backward scattering and laser mode splitting. Detecting the beat note signals of splitting variation.	0.5, 0.8, 0.3, and 20 ppb for CO ₂ , NH ₃ , NO ₂ , and H ₂ O	NO ₂ molecule dynamics process and multicomponent label-free sensing	[16]
	H ¹³ CN	Two silica micro-disk resonator	3×10^8	Molecular absorption spectra of dual comb	–	Miniature size	[187]

6. Sensing applications in a wider spectrum range: Gas molecule absorption spectra are primarily concentrated in the mid-infrared frequency range [199]. Utilizing microcavities for integrated and cost-effective gas detection in this spectral range presents a significant challenge and opportunity.

7. Molecular dynamics and cellular evolution: The micro-nanoscale structure and high sensitivity of microcavities provide a unique advantage for exploring molecular dynamics [174] and cellular evolution [151] within cells.

8. Development of stable, miniaturized, and mass-produced microcavities: Various applications, including environmental monitoring and point-of-care testing, necessitate long-term operation, high reliability, and stability [133]. Achieving scalability, low cost, and miniaturization of devices is essential in these scenarios [17].

Acknowledgments

Funding: The authors acknowledge the support from the National Key Research and Development Program of China (2023YFB2806200 and 2023YFB2805600), the National Natural Science Foundation of China (grants 61975025 and U2130106), the Key Project of China Southern Power Grid (CG2100022001608777), and the China Postdoctoral Science Foundation (grant BX20220056).

Author contributions: All the authors have accepted responsibility for the entire content of this submitted manuscript and approved the submission.

Competing interests: The authors declare that they have no competing interests.

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Optical Microcavities Empowered Biochemical Sensing: Status and Prospects

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Citation: Guo Y, Liang Y, Li Y, Tian B, Fan X, He Y, Liu M, Peng L, Tang N, Tan T, et al. Optical Microcavities Empowered Biochemical Sensing: Status and Prospects. *Adv Devices Instrum.* 2024;5:0041. DOI: 10.34133/adi.0041

Optical microcavities are compact structures that confine resonant photons in microscale dimensions for long periods of time, greatly enhancing light–matter interactions. Plentiful and profound physical mechanisms within these microcavities or functional microcavities have been extensively explored, including mode shift/splitting/broadening, lasing and gain enhancements, surface plasmon resonance, fluorescence resonance energy transferring, optical frequency comb spectroscopy, optomechanical interaction, and exceptional point. The versatility in design and the diverse range of materials, particularly composites involving metals and 2-dimensional materials, have paved a way for innovative approaches and improved performance in biochemical sensing applications. Leveraging the advantages ranging from miniaturization, high sensitivity, rapid response, and inherent stability, optical microcavity-based biochemical sensors have emerged to address the growing and increasingly complex demands of biochemical detection. This review commences with an exploration of fundamental mechanisms and structures and then delves into typical applications in recent advancements, covering the detection of biomacromolecules, cells, solid particles, liquid ions, and gas molecules. This review also culminates with a forward-looking perspective, highlighting future development trends and crucial research directions.

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Advanced Devices & Instrumentation (ISSN 2767-9713) is published by the American Association for the Advancement of Science. 1200 New York Avenue NW, Washington, DC 20005.

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